QoS Multicast Routing Utilizing Cross-Layer Design for IoT-Enabled MANET in RIS-Aided Cell-Free Massive MIMO

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***Abstract*—This article proposes a novel QoS multicast routing (QSMR) protocol that leverages cross-layer design for IoT-enabled mobile ad hoc networks (MANETs) within a reconfigurable intelligent surface (RIS)-aided cell-free massive MIMO (CF-mMIMO) environment, especially when multiple active eavesdroppers (Eves) are present. The proposed QSMR protocol employs a cross-layer approach to establish a QoS multicast mesh (QMM) route, considering spectrum efficiency (SE), achievable secrecy rate (ASR), end-to-end (E2E) delay, route stability, and hop count requirements. Initially, we formulate the SE expressions for each IoT device in both uplink and downlink, and the downlink ASR expressions for users under attack. Second, we propose two optimization problems: 1) the phase-shift control (PSC) problem, aimed at maximizing the minimum uplink SE and 2) the power control (PC) problem, focused on maximizing the minimum ASR. Additionally, a deep neural network is designed to reduce the computation time required for solving the PSC and PC problems. Third, leveraging the cross-layer synergy, we propose the QSMR protocol to establish the QMM route from a source to multiple destinations by seamlessly integrating information from both the physical and network layers. Finally, the simulation results evaluate the secure performance and provide comparisons for different scenarios. Moreover, in the presence of Eves, the proposed QSMR protocol outperforms other routing protocols in terms of routing delay, control overhead, and packet delivery ratio.**

***Index Terms*—Cell-free massive MIMO (CF-mMIMO), cross-layer, deep neural network (DNN), IoT-enabled mobile ad hoc network (MANET), optimization, Quality-of-Service multicast routing (QSMR), reconfigurable intelligent surface (RIS), secrecy rate, spectrum efficiency (SE).**

I. Introduction

CELL-FREE massive MIMO (CF-mMIMO) is a groundbreaking technology that enhances spectrum efficiency (SE), expands coverage, and reduces user interference. This is achieved by dispersing numerous access points (APs) across a vast region, deviating from the conventional fixed cell structures of traditional networks [1], [2]. The reconfigurable intelligent surface (RIS) was introduced as an innovative technique to enhance wireless network performance by actively altering the path of electromagnetic waves to boost signal strength and expand coverage [3], [4]. Furthermore, physical layer security (PLS) augments security by exploiting the unique attributes of wireless channels, particularly in contexts vulnerable to data interception [5], [6]. The convergence of PLS, RIS, and CF-mMIMO technologies has garnered interest as a means to enhance wireless communications in 5G, IoT, and smart cities [7], [8], [9], [10], [11]. QoS multicast routing (QSMR) has emerged as a pivotal technology, enabling the transmission of data packets from a single source to multiple recipients while upholding specific Quality-of-Service (QoS) standards. This is crucial for satisfying QoS demands in wireless networks [12], [13], [14]. Such routing is widely utilized in multimedia streaming, video conferencing, and online gaming, where data dissemination to multiple users concurrently is essential.

A. Related Work

Timilsina et al. [7] investigated the achievable secrecy rates (ASRs) within a CF-mMIMO system featuring an eavesdropper (Eve). They formulated a power allocation algorithm to optimize the ASR, underscoring the significant rate potentially exposed to the Eve. Zhang et al. [8] introduced a secure communication framework for an RIS-aided CF-mMIMO system equipped with low-resolution ADCs/DACs, considering active Eves. This framework utilized conjugate/random beamforming and a minimum mean-squared-error channel estimator, culminating in a closed-form expression for the achievable ergodic secrecy rate. The study in [11] proposed a security mechanism and addressed optimization challenges using path-following algorithms. Their approach aimed to maximize data and secrecy rates while minimizing power consumption in CF-mMIMO networks, providing closed-form solutions. Xu et al. [15] showcased a novel QoS-aware secure routing methodology for multihop wireless networks. This method employed a distinctive theoretical framework, leveraging Bellman–Ford and Dijkstra's algorithms for effective route selection, while countering threats from malicious Eves and self-centered jammers. Safari et al. [16] introduced a notable routing protocol named cross-layer adaptive fuzzy-based ad hoc on-demand distance vector (CLAF-AODV).

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This protocol harnessed a two-tier fuzzy-logic system and cross-layer design, aiming to intelligently pinpoint nodes with elevated broadcasting probabilities. It considered parameters from the initial three layers of the OSI model, with objectives to curtail broadcast packets, elevate throughput, diminish packet loss, and economize energy. Finally, [17] presented the hybrid genetic firefly algorithm-based routing protocol (HGFA), which combined the genetic algorithm's traits with the firefly algorithm, catering to swift and dependable VANET communication. A comprehensive comparative assessment affirmed HGFA's superiority, evidencing marked performance enhancements over both firefly and PSO techniques in varied network densities.

B. Motivations

In the rapidly evolving communication landscape, characterized by technological advancements and escalating user demands, there exists a pressing need for secure and dependable communication, particularly within 5G and IoT networks. The imperative to implement QSMR with a cross-layer design in IoT-enabled mobile ad hoc networks (MANETs), supported by RIS-aided CF-mMIMO, is clear. This approach enhances the quality, security, and dependability of group communication. While the studies referenced in Section I-A present limitations, other pertinent research also reveals discernible gaps.

1) Most studies have focused on configurations with a single antenna and one Eve. Such systems typically operate under specific conditions and are devoid of RIS support [7], [8], [11]. Moreover, while prevailing research aims to optimize SE and ASR, these strategies may not adequately cater to the QoS needs of all user equipments (UEs). Hence, addressing the max–min fairness challenge is vital to ensure equitable resource allocation, subsequently enhancing QoS for all UEs.
2) Earlier security research focused on enhancing the ASR but often overlooked the simultaneous consideration of both SE and ASR for IoT devices [7], [8], [11].
3) A discernible research void exists concerning QSMR with a cross-layer design, especially tailored for IoT-enabled MANETs in an RIS-aided CF-mMIMO environment.
4) Studies on QoS routing in distributed contexts [15], [16], [17] predominantly rely on the tenets of ad hoc on-demand distance vector (AODV) [18] and multicast AODV (MAODV) [19]. They integrate emerging technologies to discern when they relay or discard route request (RREQ) packets during route discovery. Such research primarily seeks to forge either a singular route or a tree structure, prompting our exploration into the merits of mesh routing, which promises enhanced stability, reduced rerouting necessities, and augmented system performance.

Bridging these research gaps is pivotal for the future trajectory of trustworthy communication systems.

C. Contributions

One of the pivotal innovations in our research is the integration of cross-layer synergy. This approach is not merely a design choice but a strategic decision to utilize the strengths of both the physical and network layers, leading to a more robust and efficient system. Cross-layer synergy allows for a seamless flow of information between the layers, ensuring that each layer is aware of the conditions and requirements of the other. This mutual awareness results in a dynamic system that can adapt to changing conditions in real time. Furthermore, by leveraging cross-layer synergy, our system can preemptively address potential challenges, enhancing overall system performance. In response to the challenges in the motivations, this article describes the following principal contributions.

1) We introduce a groundbreaking QSMR protocol to forge a QoS multicast mesh (QMM) route via a cross-layer strategy, addressing QoS criteria, such as SE, ASR, end-to-end (E2E) delay, route stability, and hop count.
2) Initially, we extract closed-form expressions for channel estimates and SEs for IoT devices across both uplink and downlink. We also derive expressions for downlink ASRs for compromised IoT devices, employing the MMSE combining technique.
3) We then introduce two innovative max–min fairness optimization challenges for IoT devices: a) a phase-shift control (PSC) challenge centered on optimizing the minimum uplink SEs and b) a power control (PC) challenge aimed at maximizing the minimum downlink ASRs. Concurrently, a deep neural network (DNN) is architected to truncate the computational duration needed to address the PSC and PC challenges.
4) Subsequently, we advocate for the QSMR protocol to craft the QMM route from a singular source to multiple endpoints, harnessing data from both the physical and network strata. This data spans SEs, ASRs, position, direction, E2E delay, route stability, and hop count.
5) In conclusion, we assess security efficacy via simulation outcomes and draw comparisons across diverse scenarios. Notably, the QSMR protocol manifests a marked edge over alternative routing protocols, especially in metrics, such as routing delay, control overhead, and packet delivery ratio (PDR) in the presence of Eves.

D. Notations

The imaginary unit of a complex number is denoted by $j = \sqrt{-1}$. Superscripts ^{T}, *, and ^{H} denote the transpose, conjugate, and conjugate transpose, respectively. The notations $[\mathbf{A}]_{n:}$ and $[\mathbf{A}]_{:m}$ denote the nth row and mth column of a matrix $\mathbf{A}$, respectively. A diagonal matrix with diagonal entries $(a_1, \ldots, a_n)$ is denoted by $\mathrm{diag}(a_1, \ldots, a_n)$. A block-diagonal matrix is denoted by $\mathrm{diag}(\mathbf{A}_1, \ldots, \mathbf{A}_n)$, where the diagonal entries are square matrices $\mathbf{A}_1, \ldots, \mathbf{A}_n$. The rank of a matrix $\mathbf{A}$ is denoted by $\mathrm{rank}(\mathbf{A})$. The product of two real numbers M and N is denoted by MN. Table I introduces the summary of notations used throughout this article.

II. System Model

We examine an IoT-enabled MANET within an RIS-aided CF-mMIMO system, as depicted in Fig. 1. This system comprises a central processing unit (CPU), K single-antenna IoT

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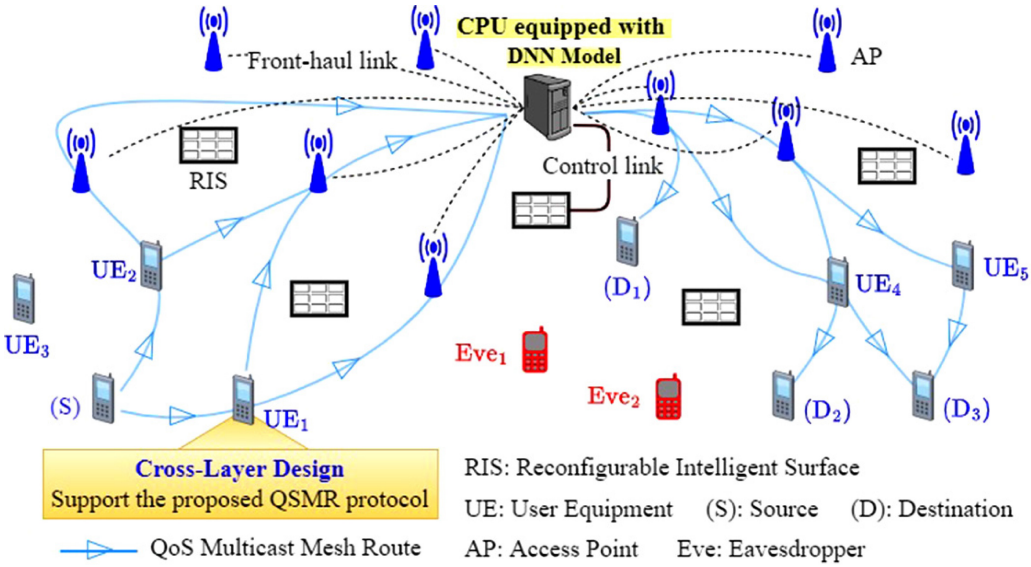

Fig. 1. Illustration of QSMR for IoT-enabled MANET in RIS-aided CF-mMIMO: basic concepts.

TABLE I
MATHEMATICAL NOTATIONS

Symbol	Description
L, R, K	n. of APs, RISs, UEs
N, M	n. of AP antennas, RIS's elements
τ_c, τ_p	TDD frame length, pilot training length
$\tau_{\mathrm{ul}}, \tau_{\mathrm{dl}}$	UL, DL data trans. length
$\boldsymbol{\phi}_r, \boldsymbol{\Phi}_r$	response vector, response matrix of RIS$_r$
$\mathbf{h}_{lk}$	channel between UE$_k$ and AP$_l$
$\mathbf{f}_{rk}$	channel between UE$_k$ and RIS$_r$
f_{rmk}	channel betwwen UE$_k$ and m-th element of RIS$_r$
$\mathbf{G}_{lr}$	channel between RIS$_r$ and AP$_l$
$\mathbf{g}_{lrm}$	channel between m-th element of RIS$_r$ and AP$_l$
$\boldsymbol{\mathcal{G}}_{lrmk}$	cascaded channel between UE$_k$ and AP$_l$ through m-th element of RIS$_r$
$\boldsymbol{\mathcal{G}}_{0,lrmk}$	channel $\boldsymbol{\mathcal{G}}_{lrmk}$ with phase-shift zero
$\boldsymbol{\mathcal{G}}_{lrk}$	cascaded channel between UE$_k$ and AP$_l$ through RIS$_r$
$\widehat{\boldsymbol{\mathcal{G}}}'_{lrk}$	the matrix $\left(\widehat{\boldsymbol{\mathcal{G}}}_{0,lr1k}, \ldots, \widehat{\boldsymbol{\mathcal{G}}}_{0,lrMk}\right)$
$\boldsymbol{\mathcal{F}}_{lrmk}$	aggregate (agg.) channel between UE$_k$ and AP$_l$ through m-th element of RIS$_r$
$\boldsymbol{\mathcal{F}}_{0,lrmk}$	agg. channel $\boldsymbol{\mathcal{F}}_{lrmk}$ with phase-shift zero
$\boldsymbol{\mathcal{F}}_{lrk}$	agg. channel between UE$_k$ and AP$_l$ through RIS$_r$
$\boldsymbol{\mathcal{F}}_{l\mathcal{R}k}$	agg. channel between UE$_k$ and AP$_l$ through all RIS$_r$
$\widetilde{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k}$	channel estimation error
$\mathbf{R}^{\star}$	spatial correlation matrix of channel $\star$
$\widehat{\star}$	estimation of channel $\star$
$\boldsymbol{\varphi}(k)$	the pilot sequence used by UE$_k$
ζ_i	i-th sub-phase in channel estimation phase
$\sqrt{\eta_i}$	transmit power of UE$_i$
$(\mathrm{UE}_i, \mathrm{UE}_k)^{\mathbf{dir}}$	direct link between UE$_i$ and UE$_k$
$(\mathrm{UE}_i, \mathrm{UE}_k)^{\mathbf{cas}}$	cascaded link $(\mathrm{UE}_i, \{\mathrm{AP}\}, \mathbf{CPU}, \{\mathrm{AP}\}, \mathrm{UE}_k)$
$\mathrm{DL}(i,k)$	time taken to send a packet from UE$_i$ to UE$_k$
$\mathrm{DL}(\mathcal{R})$	End-to-End delay of the route $\mathcal{R}$
$\mathrm{LS}(i,k)$	link stability for direct link $(\mathrm{UE}_i, \mathrm{UE}_k)^{\mathbf{dir}}$
SE	spectrum efficiency
ASR	achievable secrecy rate
$\mathrm{HC}(\mathcal{R})$	hop count of the route $\mathcal{R}$
$\mathbf{cost}(\mathcal{R})$	QoS cost function of $\mathcal{R}$
RREQ	route request packet
RREP	route reply packet

devices, L APs each equipped with N antennas, $R = L$ RISs with M reflecting elements, and several active Eves. Each RIS exclusively serves one AP under the control of the CPU. The APs maintain a connection to the CPU through an assumed error-free and infinite capacity fronthaul network. Each IoT device operates as a UE and acts as a node within the MANET. Each node has knowledge of its location via the global positioning system (GPS) [20], [21]. It is assumed that each UE is aware of the channel state information (CSI) of its peers, facilitating direct data transmission between them. However, Eves, lacking knowledge of the UEs' CSI, cannot decode data exchanged directly between users or data transmitted from UEs to APs. Due to the standardized and public nature of the pilot sequences, the Eves can send illegal pilot sequences to all APs. This deceptive action causes the APs to mistakenly optimize the downlink for conditions that match the channel of the Eves, rather than the channel of the intended, attacked UEs. This enables the Eves to more easily decode or eavesdrop on the downlink transmissions intended for legitimate UEs, thereby compromising the system's security.

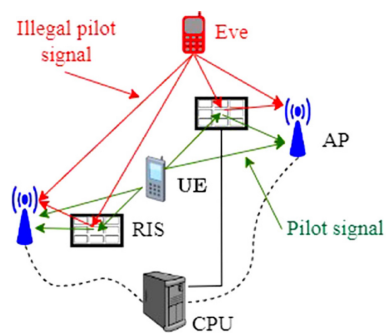

Fig. 2. Illustration of the channel estimation phase.

APs and UEs function within the physical layer, adhere to the time-division duplex (TDD) protocol [22]. This protocol segments each coherence block, denoted as τ_c, into three distinct phases: 1) τ_p for channel estimation (refer to Fig. 2); 2) τ_{ul} for uplink data transmission (see Fig. 3); and 3) τ_{dl} for downlink data transmission (illustrated in Fig. 4). The QSMR protocol leverages information from both the physical and network layers to establish the QMM route, leading from the

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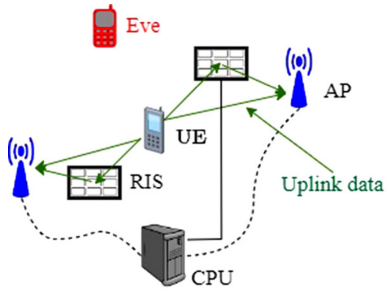

Fig. 3. Illustration of the uplink signal transmission phase.

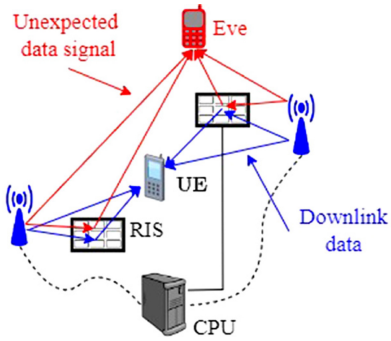

Fig. 4. Illustration of the downlink signal transmission phase.

source to the multicast group. This protocol encompasses route discovery, route reply (RREP), data transmission, and route maintenance processes. We denote the set of APs as $\mathcal{L} \triangleq \mathrm{AP}_1, \ldots, \mathrm{AP}_L$, the set of UEs as $\mathcal{K} \triangleq \mathrm{UE}_1, \ldots, \mathrm{UE}_K$, the set of RISs as $\mathcal{R} \triangleq \mathrm{RIS}_1, \ldots, \mathrm{RIS}_R$, and the set of Eves as $\mathcal{J} = \mathcal{E}_1, \ldots, \mathcal{E}_J$. In this context, L, K, R, and J represent the total numbers of APs, UEs, RISs, and Eves, respectively.

A. Illustration of the Proposed QSMR Protocol: Basic Concepts

As depicted in Fig. 1, two distinct node groups operate at a considerable distance from each other. Let S symbolize the source, and let $\mathfrak{D} = \mathrm{D}_1, \mathrm{D}_2, \mathrm{D}_3$ represent the multicast group. Assuming that S is aware of the multicast IP address, the QSMR protocol employs the following steps to establish a QMM route from S to $\mathfrak{D}$.

Initialization: Each node sets up its routing information and routing table. In the cross-layer design (refer to Fig. 5), APs and the CPU collaborate to determine the optimal phase shift for the RISs and the ideal transmit powers for the APs. This ensures the maximization of uplink SEs and downlink ASRs for all UEs, especially in the presence of Eves. To expedite computations, the CPU employs a DNN model, predicting combining vectors, optimal transmit power, SE, and ASR.

Route Discovery: To explore feasible QoS mesh routes to the multicast group, the route discovery process involves several key steps as outlined below:

Step 1: S broadcasts an RREQ packet to its neighboring nodes, appending a route cost (RC) and other routing parameters. The RC is structured as a vector encompassing SE, ASR, E2E delay, and hop count.

Step 2: APs intercept the RREQ packet from S and relay it to the CPU. The CPU then calculates the SE of (S) and ASR for all UEs, updating the RREQ's SE. Subsequently, the CPU dispatches the updated RREQ packet to all UEs.

Step 3: Upon receiving the RREQ packet from the CPU, each UE extracts the packet's information, requests their ASR data from the CPU, and updates the RREQ's RC. If the cost meets the threshold and is lower than the RC in the routing table, the UE refreshes the routing table with the new cost and the preceding node. If not, the RREQ is discarded.

Step 4: UE_1, UE_2, and UE_3 also intercept an RREQ packet from S and replicate the actions from step 3. Assuming the new RCs at UE_3 do not align with the threshold, all RREQ packets are discarded.

Step 5: UE_1, UE_2, UE_4, and UE_5 iterate through steps 1–3 until a destination node receives the RREQ packet. The destinations in $\mathfrak{D}$ then initiate the RREP process.

RREP: Following the successful identification of feasible routes, this process encompasses the steps involved in responding back to the source node with the selected route information.

Step 6: After receiving the RREQ from S, each D_i computes a fresh RC. If this new RC aligns with the threshold conditions, D_i dispatches an RREP packet back to its preceding UEs. If not, D_i discards the RREQ. It is worth noting that each destination D_i can accommodate multiple routes from varying neighbors that meet the threshold, culminating in the formation of a mesh route.

Step 7: Any UE that receives an RREP from a destination continues to forward the RREP. This process persists until S receives RREPs from all destinations, solidifying the QMM route that includes S, UE_1, UE_2, UE_4, UE_5, D_1, D_2, and D_3.

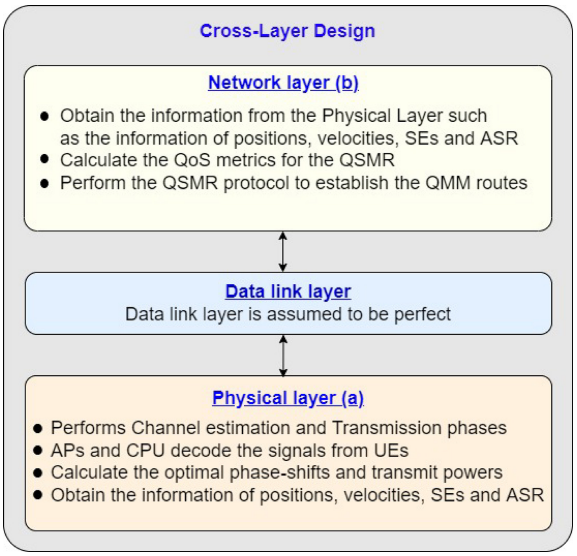

Fig. 5. Cross-layer design for QSMR protocol: basic concept.

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B. Cross-Layer Design for the Proposed QSMR Protocol

In our cross-layer design approach, the data link layer is presumed to be flawless, with our primary emphasis placed on the physical and network layers. Information from the physical layer is channeled to the network layer, aiding the proposed QSMR protocol in crafting the QMM route from a source to the multicast group. The delineated roles of the physical and network layers within the cross-layer design, as depicted in Fig. 5, are elucidated as follows.

1) Physical Layer: The physical layer process encompasses the channel estimation phase, detailed in Section III titled "Channel Estimation Phase in Physical Layer," and the transmission phases, detailed in Section IV titled "Transmission Phases in Physical Layer." In physical layer (a) of Fig. 5, the following tasks are executed to gather information on uplink SE, downlink ASR, position, and velocity.

1) APs compute channel estimates for the transmission phase. These estimates assist the APs and CPU in decoding the uplink signal, transmitting the downlink signal, and computing the uplink SE and downlink ASR.
2) Using channel estimates, algorithms, and the DNN model, both the APs and CPU decode the UE's signals. They then select optimal phase shifts to maximize the minimum strength of aggregate channel estimates from the UEs. Subsequently, the APs use the MMSE combining method to enhance the received signal at the CPU. Ultimately, the CPU calculates the uplink SEs for all UEs.
3) With the channel estimates and the DNN model, the CPU determines optimal transmit powers for the APs to maximize the minimum ASR for the UEs. It also calculates the ASRs for all UEs.
4) The UEs use GPS technology to determine their positions and velocities.

2) Network Layer: The network layer employs the proposed QSMR to establish the QMM route from a source to the multicast group.

1) In the network layer (b) of Fig. 5, the UEs retrieve information on SE, ASR, position, and velocity from the physical layer. Additionally, UEs also gather details about E2E delay, route stability, and hop count from the network layer.
2) The QoS metrics, which include SE, ASR, E2E delay, route stability, and hop count, are used in the QSMR.
3) In the proposed QSMR protocol, each node determines the QMM route from a source to multiple destinations. The proposed QSMR protocol is further detailed in Section VI.

3) Routing Protocol Integration With RIS-Aided CF-MIMO Physical Layer Design: The RIS-aided CF-MIMO design provides dynamic reconfigurability, allowing the physical layer to adapt to changing network conditions. This adaptability is harnessed by the QSMR protocol to make informed routing decisions. Specifically:

1) *RIS-Aided CF-MIMO Design:* The RIS-aided component extends the coverage range and amplifies signal strength, while the CF-MIMO aspect, bolstered by a dense deployment of APs, ensures robust signal reception and transmission. Together, they create a seamless network experience. This synergy ensures stable routes, reducing the need for frequent route reestablishments, leading to decreased routing delay and control overhead. As a result, there is a marked improvement in PDR and consistent adherence to QoS requirements.
2) The RIS-aided CF-MIMO design is inherently dynamic, swiftly adapting to changing network conditions and user needs. This is due to its ability to real-time reconfigure parameters like RIS phase shifts and MIMO beamforming vectors. These adjustments are vital for addressing variations in user demands, channel conditions, and interference. This dynamic nature allows for optimized signal quality and coverage without manual tweaks. Recognizing this, the QSMR protocol leverages this adaptability to make on-the-fly routing decisions, ensuring consistent performance in varying conditions.

This integration ensures that the proposed QSMR protocol is not only aware of the physical layer's capabilities but also actively utilizes them to enhance the overall network performance.

C. RIS Configuration

We assume that each RIS includes M reflecting elements. A response vector of RIS_r is defined by

$$\boldsymbol{\phi}_r \triangleq \left(\phi_{r,1}, \phi_{r,2}, \ldots, \phi_{r,M}\right), \text{ with } \phi_{r,m} \triangleq \alpha e^{j\theta_{r,m}} \tag{1}$$

where $\alpha = 1$ is the reflection amplitude of each RIS, and each $\theta_{r,m} \in [0, 2\pi)$ is the phase shift of the mth element. A response matrix of RIS_r is defined by

$$\boldsymbol{\Phi}_r \triangleq \text{diag}\left(\phi_{r,1}, \phi_{r,2}, \ldots, \phi_{r,M}\right) \in \mathbb{C}^{M\times M}. \tag{2}$$

D. Channel Models

We denote $\mathbf{h}_{lk}$, $\mathbf{f}_{rk}$, and $\mathbf{G}_{lr}$ as the channel between UE_k and AP_l, between UE_k and RIS_r, and between RIS_r and AP_l, respectively. The channels $\mathbf{h}_{lk}$ and $\mathbf{f}_{rk}$ can be modeled by spatial correlated fading channels as follows:

$$\mathbf{h}_{lk} \sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{R}_{lk}^{\mathbf{h}}\right), \ \mathbf{f}_{rk} \sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{R}_{rk}^{\mathbf{f}}\right). \tag{3}$$

where $\mathbf{R}_{lk}^{\mathbf{h}}$ and $\mathbf{R}_{rk}^{\mathbf{f}}$ are the positive semi-definite correlation matrices that describe the large-scale fading, including geometric path loss, shadowing, antenna gains, and spatial channel correlation. The channel $\mathbf{G}_{lr}$ is modeled using the Kronecker model as follows:

$$\mathbf{G}_{lr} = (\mathbf{S}_{lr})^{1/2}\mathbf{W}_{lr}(\mathbf{T}_{lr})^{1/2} \tag{4}$$

where each element of $\mathbf{W}_{lr} \in \mathbb{C}^{N\times M}$ is an independent and identically distributed (i.i.d.) standard complex Gaussian random variable; $\mathbf{S}_{lr} \in \mathbb{C}^{N\times N}$ and $\mathbf{T}_{lr} \in \mathbb{C}^{M\times M}$ are the received Hermitian matrix for AP_l and the transmit Hermitian matrix for RIS_r, respectively. Moreover, the matrices $\mathbf{S}_{lr}$ and $\mathbf{T}_{lr}$ represent the large-scale fading for AP_l and RIS_r, respectively, which includes factors, such as geometric path loss, shadowing, antenna gains, and spatial channel correlation.

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Consequently, the channel between UE_k and the mth element of RIS_r is given by

$$f_{rmk} \triangleq [\mathbf{f}_{rk}]_m \in \mathbb{C} \tag{5}$$

and the channel between the mth element of RIS_r and AP_l is given by

$$\mathbf{g}_{lrm} \triangleq [\mathbf{G}_{lr}]_{:m} \in \mathbb{C}^N. \tag{6}$$

The cascaded channel between UE_k and AP_l through the mth element of RIS_r is given by

$$\begin{aligned} &\mathcal{G}_{lrmk} \triangleq \mathbf{g}_{lrm}[\mathbf{\Phi}_r]_{mm} f_{rmk} = \mathbf{g}_{lrm} f_{rmk} \phi_{r,m} \\ &\text{and } \mathcal{G}_{0,lrmk} \triangleq \mathbf{g}_{lrm} f_{rmk} \end{aligned} \tag{7}$$

where $\mathcal{G}_{0,lrmk}$ implies the cascaded channel $\mathcal{G}_{lrmk}$ with phase-shift zero ($\phi_{r,m} = 1$). Hence, the cascaded channel between UE_k and AP_l through an RIS_r is given by

$$\mathcal{G}_{lrk} \triangleq \sum_{m=1}^{M} \mathcal{G}_{lrmk} = \sum_{m=1}^{M} \mathcal{G}_{0,lrmk} \phi_{r,m}. \tag{8}$$

The aggregate channel between UE_k and AP_l through the mth element of RIS_r with phase-shift zero is given by

$$\mathcal{F}_{0,lrmk} \triangleq \mathbf{h}_{lk} + \mathcal{G}_{0,lrmk}. \tag{9}$$

Hence, the aggregate channel between UE_k and AP_l through the RIS_r is given by

$$\mathcal{F}_{lrk} \triangleq \mathbf{h}_{lk} + \mathcal{G}_{lrk} \tag{10}$$

and the aggregate channel between UE_k and AP_l through all the RISs is given by

$$\mathcal{F}_{l\mathcal{R}k} \triangleq \mathbf{h}_{lk} + \sum_{r=1}^{R} \mathcal{G}_{lrk}. \tag{11}$$

Lemma 1: The distributions of channels $\mathbf{g}_{lrm}$, $\mathcal{G}_{0,lrmk}$, and $\mathcal{F}_{0,lrmk}$ are given by

$$\begin{aligned} &\mathbf{g}_{lrm} \sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{R}^{\mathbf{g}}_{lrm}\right),\ \mathcal{G}_{0,lrmk} \sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{R}^{\mathcal{G}}_{lrmk}\right) \\ &\text{and } \mathcal{F}_{0,lrmk} \sim \mathcal{N}_{\mathbb{C}}\left(0, \mathbf{R}^{\mathcal{F}}_{lrmk}\right) \end{aligned} \tag{12}$$

where

$$\begin{aligned} &\mathbf{R}^{\mathbf{g}}_{lrm} = [\mathbf{T}_{lr}]_{mm} \mathbf{S}_{lr},\ \mathbf{R}^{\mathcal{G}}_{lrmk} = \mathbf{R}^{\mathbf{g}}_{lrm} \left[\mathbf{R}^{\mathbf{f}}_{rk}\right]_{mm} \\ &\mathbf{R}^{\mathcal{F}}_{lrmk} = \mathbf{R}^{\mathbf{h}}_{lk} + \mathbf{R}^{\mathbf{g}}_{lrm} \left[\mathbf{R}^{\mathbf{f}}_{rk}\right]_{mm}. \end{aligned} \tag{13}$$

Proof: The proof is given in the Appendix. ■

III. Channel Estimation Phase in Physical Layer

In this phase, APs derive channel estimates tailored for the transmission phase of physical layer signals. These estimates equip the APs and CPU to decode the uplink signal, facilitate downlink data transmission, and compute both the uplink SE and downlink ASR. By leveraging channel estimates, specific algorithms, and the DNN model, the APs and CPU collaboratively pinpoint optimal phase shifts, aiming to bolster the aggregate channel estimates originating from the UEs. We operate under the assumption that there are τ_p mutually orthogonal pilots are available, each spanning a length of τ_p samples. Drawing inspiration from [23], we posit that $\tau_p = (RM + 2)K$. Let $\boldsymbol{\varphi}(k) \in \mathbb{C}^{\tau_p}$ represent the pilot sequence utilized by UE_k, with the condition $|\boldsymbol{\varphi}(k)|^2 = 1\ \forall k \in \mathcal{K}$. Throughout this phase, UEs simultaneously transmit their distinct pilots to all APs. Concurrently, the Eves dispatch illicit pilot sequences to all APs, aiming to discern the signal directed to UEs during downlink transmission. We operate under the premise that the CPU possesses the capability to identify the presence of Eves using established techniques [11], [24], [25]. The training phase is segmented into $RM + 2$ subphases, labeled as $\zeta_0, \zeta_1, \ldots, \zeta_{RM+1}$. For simplification, we represent $\zeta_{r,m}$ as ζ_i when $i = (r-1)M + m$. For the sake of argument, we assume $\varphi_u^{\mathcal{E}} = \varphi_u$, where $\varphi_u^{\mathcal{E}}$ denotes the pilot sequence of Eve $\mathcal{E}_u$. This implies that $\mathcal{E}_u$ employs pilot φ_u to target UE_u, applicable for all $u \in \mathcal{J}$.

A. First Subphase ζ_0

During this phase, all RISs are deactivated in order to estimate all direct channels $\mathbf{h}_{lk}$ for all $k \in \mathcal{K}, l \in \mathcal{L}$. The received signal at AP_l is given by

$$\mathbf{y}_l^p = \sum_{i=1}^{K} \sqrt{\tau_p \eta_p} \mathbf{h}_{li} \varphi_i^{\mathrm{T}} + \sum_{u=1}^{J} \sqrt{\tau_p \eta_p} \mathbf{h}_{l\mathcal{E}_u} \left(\varphi_u^{\mathcal{E}}\right)^{\mathrm{T}} + \mathbf{n}_l^p \tag{14}$$

where $\mathbf{n}_l^p \in \mathbb{C}^{N \times \tau_p}$ is the thermal noise whose elements are i.i.d. $\mathcal{N}_{\mathbb{C}}(0, \sigma^2)$; and η_p is the transmit power for each pilot sequence. Every AP_l computes $\mathbf{z}_{lk}^p \triangleq \mathbf{y}_l^p \varphi_k^*$ which is given by

$$\mathbf{z}_{lk}^p = \begin{cases} \sqrt{\tau_p \eta_p} \mathbf{h}_{lk} + \mathbf{n}_l^p \varphi_k^*, & k \notin \mathcal{J} \\ \sqrt{\tau_p \eta_p} \left(\mathbf{h}_{lk} + \mathbf{h}_{l\mathcal{E}_k}\right) + \mathbf{n}_l^p \varphi_k^*, & k \in \mathcal{J} \end{cases} \tag{15}$$

where $\mathbf{n}_l^p \varphi(k)^* \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ is the resulting noise. The MMSE estimate of $\mathbf{h}_{lk}$ given $\mathbf{z}_{lk}^p$ is

$$\widehat{\mathbf{h}}_{lk} = \sqrt{\tau_p \eta_p} \mathbf{R}_{lk}^{\mathbf{h}} \left(\mathcal{C}_{lk}^{\mathbf{h}}\right)^{-1} \mathbf{z}_{lk}^p \in \mathbb{C}^N \tag{16}$$

where

$$\mathcal{C}_{lk}^{\mathbf{h}} = \begin{cases} \tau_p \eta_p \mathbf{R}_{lk}^{\mathbf{h}} + \sigma^2 \mathbf{I}_N, & k \notin \mathcal{J} \\ \tau_p \eta_p \left(\mathbf{R}_{lk}^{\mathbf{h}} + \mathbf{R}_{l\mathcal{E}_k}^{\mathbf{h}}\right) + \sigma^2 \mathbf{I}_N, & k \in \mathcal{J}. \end{cases} \tag{17}$$

Moreover, we can obtain

$$\widehat{\mathbf{h}}_{l\mathcal{E}_k} = \mathbf{R}_{l\mathcal{E}_k}^{\mathbf{h}} \left(\mathbf{R}_{lk}^{\mathbf{h}}\right)^{-1} \widehat{\mathbf{h}}_{lk}. \tag{18}$$

B. Phase $\zeta_{(r,m)}, (r, m) \in \mathcal{R} \times \mathcal{M}$

To estimate the channel $\mathcal{G}_{0,lrmk}$, we only activate the mth element of RIS_r with a phase shift of $\phi_{r,m} = 1$, while all other elements of all RISs remain OFF. The received signal at AP_l is given by

$$\begin{aligned} \mathbf{y}_{0,lrm}^p = &\sum_{i=1}^{K} \sqrt{\tau_p \eta_p} \mathcal{F}_{0,lrmi} \varphi_i^{\mathrm{T}} \\ &+ \sum_{j=1}^{J} \sqrt{\tau_p \eta_p} \mathcal{F}_{lrm\mathcal{E}_j} \left(\varphi_j^{\mathcal{E}}\right)^{\mathrm{T}} + \mathbf{n}_l^p \in \mathbb{C}^{N \times \tau_p} \end{aligned} \tag{19}$$

where $\mathcal{F}_{0,lrmi}$ is the aggregate channel (9), $\mathbf{n}_l^p \in \mathbb{C}^{N \times \tau_p}$ is the thermal noise whose elements are i.i.d. $\mathcal{N}_{\mathbb{C}}(0, \sigma^2)$; and

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η_p is the transmit power for each pilot sequence. Every AP_l computes $\mathbf{z}^p_{0,lrmk} \triangleq \mathbf{y}^p_{0,lrm}\varphi^*_k$ which is given by

$$\mathbf{z}^p_{0,lrmk} = \begin{cases} \sqrt{\tau_p\eta_p}\boldsymbol{\mathcal{F}}_{0,lrmk} + \mathbf{n}^p_l\varphi^*_k, & k \notin \mathcal{J} \\ \sqrt{\tau_p\eta_p}\left(\boldsymbol{\mathcal{F}}_{0,lrmk} + \boldsymbol{\mathcal{F}}_{0,lrm\mathcal{E}_k}\right) + \mathbf{n}^p_l\varphi^*_k, & k \in \mathcal{J}. \end{cases} \tag{20}$$

where $\mathbf{n}^p_l\varphi(k)^* \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2\mathbf{I}_N)$ is the resulting noise. Based on Lemma 1, the MMSE estimate of $\boldsymbol{\mathcal{F}}_{0,lrmk}$ given $\mathbf{z}^p_{0,lrmk}$ is computed by

$$\widehat{\boldsymbol{\mathcal{F}}}_{0,lrmk} = \sqrt{\tau_p\eta_p}\mathbf{R}^{\mathcal{F}}_{lrmk}\left(\mathcal{C}^{\mathcal{F}}_{lrmk}\right)^{-1}\mathbf{z}^p_{0,lrmk} \tag{21}$$

where

$$\mathcal{C}^{\mathcal{F}}_{lrmk} = \begin{cases} \tau_p\eta_p\mathbf{R}^{\mathcal{F}}_{lrmk} + \sigma^2\mathbf{I}_N, & k \notin \mathcal{J} \\ \tau_p\eta_p\left(\mathbf{R}^{\mathcal{F}}_{lrmk} + \mathbf{R}^{\mathcal{F}}_{lrm\mathcal{E}_k}\right) + \sigma^2\mathbf{I}_N, & k \in \mathcal{J}. \end{cases} \tag{22}$$

Moreover, we can obtain

$$\widehat{\boldsymbol{\mathcal{F}}}_{lrm\mathcal{E}_k} = \mathbf{R}^{\mathcal{F}}_{lrm\mathcal{E}_k}\left(\mathbf{R}^{\mathcal{F}}_{lrmk}\right)^{-1}\widehat{\boldsymbol{\mathcal{F}}}_{lrmk}. \tag{23}$$

Thus, the channel estimate $\hat{\mathbf{G}}_{0,lrmk}$ can be obtained by

$$\widehat{\boldsymbol{\mathcal{G}}}_{0,lrmk} = \widehat{\boldsymbol{\mathcal{F}}}_{0,lrmk} - \widehat{\mathbf{h}}_{lk}. \tag{24}$$

C. Subphase ζ_{LR+1}

In this subphase, we compute aggregate channel estimates of $\boldsymbol{\mathcal{F}}_{lrk}$ and $\boldsymbol{\mathcal{F}}_{l\mathcal{R}k}$ in (10) and (11), respectively. The channel estimate of $\boldsymbol{\mathcal{G}}_{lrk}$ in (8) for any ϕ_r is computed by

$$\widehat{\boldsymbol{\mathcal{G}}}_{lrk} = \sum_{m=1}^{M}\widehat{\boldsymbol{\mathcal{G}}}_{lrmk} = \sum_{m=1}^{M}\widehat{\boldsymbol{\mathcal{G}}}_{0,lrmk}\phi_{r,m} = \widehat{\boldsymbol{\mathcal{G}}}'_{lrk}\boldsymbol{\phi}_r \tag{25}$$

where $\widehat{\boldsymbol{\mathcal{G}}}'_{lrk} \triangleq \left(\widehat{\boldsymbol{\mathcal{G}}}_{0,lr1k}, \ldots, \widehat{\boldsymbol{\mathcal{G}}}_{0,lrMk}\right) \in \mathbb{C}^{N\times M}$ is the matrix with column vectors $\widehat{\boldsymbol{\mathcal{G}}}_{0,lrmk}$, $m = 1, \ldots, M$. The aggregate channel estimates of $\boldsymbol{\mathcal{F}}_{lrk}$ and $\boldsymbol{\mathcal{F}}_{l\mathcal{R}k}$ are given by

$$\begin{aligned} &\widehat{\boldsymbol{\mathcal{F}}}_{lrk} = \widehat{\mathbf{h}}_{lk} + \widehat{\boldsymbol{\mathcal{G}}}_{lrk} = \widehat{\mathbf{h}}_{lk} + \widehat{\boldsymbol{\mathcal{G}}}'_{lrk}\boldsymbol{\phi}_r \\ &\text{and } \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k} \triangleq \widehat{\mathbf{h}}_{lk} + \sum_{r=1}^{R}\widehat{\boldsymbol{\mathcal{G}}}_{lrk}. \end{aligned} \tag{26}$$

The channel estimation error is defined by

$$\widetilde{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k} \triangleq \boldsymbol{\mathcal{F}}_{l\mathcal{R}k} - \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k}. \tag{27}$$

Note that $\widetilde{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k}$ and $\widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k}$ are independent, and $\widetilde{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k} \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \mathbf{R}^{\widetilde{\mathcal{F}}}_{l\mathcal{R}k})$.

Based on assumptions Section II-C, assuming that RIS_r is selected to serve AP_l, CPU will choose the phase shifts of RIS_r to maximize the minimum strength of aggregate channel estimates from UEs to AP_l through RIS_r. This optimization problem is known as phase-shift selection (PSS) and can be formulated as follows:

$$\text{PSS:} \quad \max_{\boldsymbol{\phi}_r} \min_{k\in\mathcal{K}} \|\widehat{\boldsymbol{\mathcal{G}}}'_{lrk}\boldsymbol{\phi}_r + \widehat{\mathbf{h}}_{lk}\|^2 \tag{28}$$

$$\text{subject to:} \quad |\phi_{r,m}|^2 = 1, \ m = 1, \ldots, M \tag{28a}$$

where $\boldsymbol{\phi}_r \triangleq (\phi_{r,1}, \phi_{r,2}, \ldots, \phi_{r,M}) \in \mathbb{C}^M$, $\widehat{\mathbf{h}}_{lk} \in \mathbb{C}^N$ and $\widehat{\boldsymbol{\mathcal{G}}}'_{lrk} \in \mathbb{C}^{N\times M}$. Defining $\mathbf{v}_r = (\phi_{r,1}, \ldots, \phi_{r,M}, \phi_{r,M+1})^{\mathrm{T}} \in \mathbb{C}^{M+1}$, where $\phi_{r,M+1}$ is an auxiliary variable with $|\phi_{r,M+1}|^2 = 1$, and

$$\boldsymbol{\mathcal{G}}_{lk} \triangleq \begin{bmatrix} \widehat{\boldsymbol{\mathcal{G}}}'^{\mathrm{H}}_{lrk}\widehat{\boldsymbol{\mathcal{G}}}'_{lrk} & \widehat{\boldsymbol{\mathcal{G}}}'^{\mathrm{H}}_{lrk}\widehat{\mathbf{h}}_{lk} \\ \widehat{\mathbf{h}}^{H}_{lklrk} & 0 \end{bmatrix}. \tag{29}$$

The PSS problem can be rewritten by

$$\text{PSS:} \quad \max_{\mathbf{v}_r} \min_{k\in\mathcal{K}} \mathbf{v}^H_r\boldsymbol{\mathcal{G}}_{lk}\mathbf{v}_r + \|\widehat{\mathbf{h}}_{lk}\|^2 \tag{30}$$

$$\text{subject to:} \quad |\mathbf{v}_{r,m}|^2 = 1, \ m = 1, \ldots, M+1. \tag{30a}$$

We can further convert the PSS problem into

$$\text{PSS:} \quad \max_{\mathbf{v}_r, t} \ t \tag{31}$$

$$\text{subject to:} \quad \mathbf{v}^H_r\boldsymbol{\mathcal{G}}_{lk}\mathbf{v}_r + \|\widehat{\mathbf{h}}_{lk}\|^2 - t \geq 0 \ \forall k \in \mathcal{K} \tag{31a}$$

$$|\mathbf{v}_{r,m}|^2 = 1, \ m = 1, \ldots, M+1. \tag{31b}$$

The (PSS) problem is a quadratically constrained quadratic programming (QCQP) problem that can be reformulated as a semi-definite programming (SDP) problem. Therefore, the PSS can be efficiently solved using CVXPY, a Python-embedded modeling language for convex optimization problems [26] and [27].

Remark 1: In contrast to cellular networks that require channel estimation for both uplink and downlink, there is no need to estimate downlink channels in CF-mMIMO systems [2]. Therefore, the solution to the PSS problem can be applied for both the uplink and downlink.

Remark 2: The deployment of RISs offers enhanced signal quality and coverage. However, this advancement also presents challenges, one of the most significant being optimal power supply. It is essential to ensure that RISs operate at peak performance without consuming excessive power. This section delves into potential strategies and future research directions that focus on the power-efficient deployment of RISs. The goal is to ensure that they not only enhance communication capabilities but also remain energy-efficient. There are some promising directions for future studies as follows.

1) *Sustainable Phase-Shift Management:* Future approaches should focus on optimizing the phase shifts of RIS for extended periods. Identifying configurations that remain effective over time can reduce the need for frequent recalibrations, leading to significant energy savings.
2) *Intelligent PSC:* Future studies should delve deeper into refining algorithms that determine the most efficient phase shifts for RIS elements. The aim would be to maximize signal reflection while minimizing power consumption, thereby reducing the dependency on high-power transmissions.
3) *Predictive AI-Driven Adjustments:* The integration of advanced AI and machine learning techniques offers a promising avenue. These tools can predict upcoming network conditions, allowing RIS to adjust its settings proactively. Such proactive adjustments can lead to a more energy-efficient system, reducing the need for sudden, power-intensive changes.
4) *Dynamic Energy Management:* An exciting direction for research is the development of more sophisticated dynamic power management algorithms. These would enable RIS to adjust its energy consumption based on real-time network demands, ensuring that energy is used judiciously and only when necessary.

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IV. Transmission Phases in Physical Layer

A. Uplink Signal Transmission Phases

During this phase, the signals of all UEs in the process of the QSMR protocol are simultaneously transmitted to the APs. Then, APs apply the MMSE combining vectors to maximize the SE for uplink data transmission. Finally, the CPU computes the uplink SEs of all UEs for the routing process. Specifically, UE_i sends a data symbol $s_i \in \mathbb{C}$ with a power $\sqrt{\eta_i}$, $0 \le \eta_i \le \eta_{\max}$. As all APs receive the signal from all UEs, the received signal at AP_l can be expressed as

$$\mathbf{y}_l^{ul} = \sum_{i=1}^{K} \sqrt{\eta_i} \boldsymbol{\mathcal{F}}_{l\mathcal{R}i} s_i + \mathbf{n}_l \in \mathbb{C}^N \tag{32}$$

where $\mathbf{n}_l \sim \mathcal{N}_{\mathbb{C}}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$ is the independent noise. Every AP_l forwards $\mathbf{y}_l^{ul}$ to the CPU which selects the received combining vectors $\mathbf{v}_{lk} \in \mathbb{C}^N$ for UE_k to detect the data symbol s_k. The received signal at CPU can be expressed as

$$\begin{aligned} r_k^{ul} &= \sum_{l=1}^{L} \mathbf{v}_{lk}^H \left(\sum_{i=1}^{K} \sqrt{\eta_i} \boldsymbol{\mathcal{F}}_{l\mathcal{R}i} s_i + \mathbf{n}_l \right) \\ &= \sqrt{\eta_k} \left(\sum_{l=1}^{L} \mathbf{v}_{lk}^H \boldsymbol{\mathcal{F}}_{l\mathcal{R}k} \right) s_k \\ &\quad + \sum_{\substack{i=1 \\ i \neq k}}^{K} \sqrt{\eta_i} \left(\sum_{l=1}^{L} \mathbf{v}_{lk}^H \boldsymbol{\mathcal{F}}_{l\mathcal{R}i} \right) s_i + \sum_{l=1}^{L} \mathbf{v}_{lk}^H \mathbf{n}_l. \end{aligned} \tag{33}$$

We define $\boldsymbol{\mathcal{F}}_k \triangleq (\boldsymbol{\mathcal{F}}_{1k}^{\mathrm{T}} \ldots \boldsymbol{\mathcal{F}}_{Lk}^{\mathrm{T}})^{\mathrm{T}} \in \mathbb{C}^{LN}$ as the collective channel from all APs, and $\mathbf{v}_k \triangleq (\mathbf{v}_{1k}^{\mathrm{T}} \ldots \mathbf{v}_{Lk}^{\mathrm{T}})^{\mathrm{T}} \in \mathbb{C}^{LN}$ as the collective combing vector for all APs. The expression (33) can be rewritten by

$$\hat{s}_k = \sqrt{\eta_k} \mathbf{v}_k^H \boldsymbol{\mathcal{F}}_k s_k + \sum_{i=1, i \neq k}^{K} \sqrt{\eta_i} \mathbf{v}_k^H \boldsymbol{\mathcal{F}}_i s_i + \mathbf{v}_k^H \mathbf{n}_k' \in \mathbb{C} \tag{34}$$

where $\mathbf{n}_k' \triangleq (\mathbf{n}_1^{\mathrm{T}} \ldots \mathbf{n}_L^{\mathrm{T}})^{\mathrm{T}}$. The instantaneous SINR of r_k^{ul} can be computed by

$$\mathrm{SINR}_k^{ul} = \frac{\eta_k \left\| \mathbf{v}_k^H \widehat{\boldsymbol{\mathcal{F}}}_k \right\|^2}{\sum_{i=1, i \neq k}^{K} \eta_i \left\| \mathbf{v}_k^H \widehat{\boldsymbol{\mathcal{F}}}_i \right\|^2 + \mathbf{v}_k^H \mathbf{Z}_k \mathbf{v}_k} \tag{35}$$

where $\mathbf{Z}_k = (\sum_{i=1}^{K} \eta_i \mathbf{R}_i^{\widetilde{\mathcal{F}}} + \sigma^2 \mathbf{I}_{LN})$ and the collection matrix of estimation errors $\mathbf{R}_i^{\widetilde{\mathcal{F}}} \triangleq \mathrm{diag}\left(\mathbf{R}_{1\mathcal{R}i}^{\widetilde{\mathcal{F}}}, \ldots, \mathbf{R}_{L\mathcal{R}i}^{\widetilde{\mathcal{F}}} \right)$. Moreover, the SINR_k^{ul} has the form of generalized Rayleigh quotient. Hence, it can be maximized by the optimal collective MMSE combining [22, Lemma B.10]

$$\mathbf{v}_{lk}^{\mathrm{MMSE}} = \eta_k \left(\sum_{i=1}^{K} \eta_i \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}i} \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}i}^H + \mathbf{Z}_k \right)^{-1} \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k} \tag{36}$$

which leads to the maximum

$$\mathrm{SINR}_k^{ul} = \eta_k \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k}^H \left(\sum_{i=1, i \neq k}^{K} \eta_i \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}i} \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}i}^H + \mathbf{Z}_k \right)^{-1} \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k}. \tag{37}$$

Thus, uplink SE for UE_k is given by

$$\mathrm{SE}_k^{ul} = \frac{\tau_{ul}}{\tau_c} \log_2 \left(1 + \mathrm{SINR}_k^{ul} \right). \tag{38}$$

B. Downlink Signal Transmission Phase

In this phase, each AP uses the channel estimates and the conjugate beamforming to transmit signals to all UEs in the process of the QSMR protocol. Moreover, the optimal powers of APs are selected to maximize the minimum ASR of UEs in the presence of Eves. Finally, the CPU computes the ASR of all UEs for the routing process. Let $s_i \in \mathbb{C}$ be the data symbol for UE_i and η_{li} be the transmit power of AP_l to UE_i, where $\mathbb{E}\{|s_i|^2\} = 1$ and $0 \le \eta_{li} \le \eta_{\max}$. The transmitted signal from AP_l is

$$x_l = \sum_{i=1}^{K} \sqrt{\eta_i} \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}i} s_i. \tag{39}$$

Hence, the received signals at UE_k and $\mathscr{E}_u$ are, respectively, given by

$$r_k^{dl} = \sum_{l=1}^{L} \sum_{i=1}^{K} \boldsymbol{\mathcal{F}}_{l\mathcal{R}k}^H \sqrt{\eta_i} \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}i} s_i + n_k^{dl} \tag{40}$$

$$r_{\mathscr{E}_u}^{dl} = \sum_{l=1}^{L} \sum_{i=1}^{K} \boldsymbol{\mathcal{F}}_{lR\mathscr{E}_u}^H \sqrt{\eta_i} \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}i} s_i + n_{\mathscr{E}_u}^{dl} \tag{41}$$

where n_k^{dl} and $n_{\mathscr{E}_u}^{dl}$ are the thermal noises which are i.i.d. $\mathcal{N}_{\mathbb{C}}(0, \sigma_{dl}^2)$. Similar to (33) and (34), the SINR of r_k^{dl} is given by

$$\mathrm{SINR}_k^{dl} = \frac{\eta_k \left| \mathbb{E}\{\boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_k\} \right|^2}{\sum_{i=1}^{K} \eta_i \mathbb{E}\left\{ \left| \boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_i \right|^2 \right\} - \eta_k \left| \mathbb{E}\{\boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_k\} \right|^2 + \sigma_{dl}^2}. \tag{42}$$

By substituting $\boldsymbol{\mathcal{F}}_{l\mathcal{R}k} = \widehat{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k} + \widetilde{\boldsymbol{\mathcal{F}}}_{l\mathcal{R}k}$ into expressions of (42), we have

$$\begin{aligned} \mathbb{E}\{\boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_k\} &= \mathbb{E}\{\widehat{\boldsymbol{\mathcal{F}}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_k\} \\ \mathbb{E}\left\{ \left| \boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_i \right|^2 \right\} &= \mathbb{E}\left\{ \left| \widehat{\boldsymbol{\mathcal{F}}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_i \right|^2 \right\} + \mathbb{E}\left\{ \left| \widetilde{\boldsymbol{\mathcal{F}}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_i \right|^2 \right\} \end{aligned} \tag{43}$$

where $\widetilde{\boldsymbol{\mathcal{F}}}_k \triangleq (\widetilde{\boldsymbol{\mathcal{F}}}_{1\mathcal{R}k}^{\mathrm{T}} \ldots \widetilde{\boldsymbol{\mathcal{F}}}_{L\mathcal{R}k}^{\mathrm{T}})^{\mathrm{T}} \in \mathbb{C}^{LN}$. Assuming that Eves have perfect knowledge of its own channels. Hence, based on (41), the $\mathrm{SINR}_{\mathscr{E}_u}^{dl}$ with $u \in \mathcal{J}$ can be expressed as

$$\mathrm{SINR}_{\mathscr{E}_u}^{dl} = \frac{\eta_u \mathbb{E}\left\{ \left| \boldsymbol{\mathcal{F}}_{\mathscr{E}_u}^H \widehat{\boldsymbol{\mathcal{F}}}_u \right|^2 \right\}}{\sum_{i=1, i \neq u}^{K} \eta_i \mathbb{E}\left\{ \left| \boldsymbol{\mathcal{F}}_{\mathscr{E}_u}^H \widehat{\boldsymbol{\mathcal{F}}}_i \right|^2 \right\} + \sigma_{dl}^2}. \tag{44}$$

From (42) and (44), the ASR of UE_k, with $k \in \mathcal{K}$ is given by

$$\mathrm{ASR}_k^{dl} = \left[\log_2\left(1 + \mathrm{SINR}_k^{dl}\right) - \log_2\left(1 + \mathrm{SINR}_{\mathscr{E}_k}^{dl}\right) \right]^+ \tag{45}$$

where $[\star]^+ = \max\{0, \star\}$, and if $k \notin \mathcal{J}$, $\mathrm{SINR}_{\mathscr{E}_k}^{dl} := 0$, i.e., ASR_k^{dl} becomes achievable data rate of UE_k.

Remark 3: The expressions in (43) can be computed on the CPU using the Monte Carlo method, which relies on the channel estimates $\widehat{\boldsymbol{\mathcal{F}}}_k$ and channel estimate errors $\widetilde{\boldsymbol{\mathcal{F}}}_k$. This process can be explained as follows.

CPU Calculations:

1) *Data Collection:* The CPU collects the pilot sequences sent from the UEs to estimate the CSI.

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2) *Channel Estimation:* Using the received pilot sequences, the CPU estimates the channel gains, path loss, and other relevant channel characteristics.
3) *Monte Carlo Simulation:* Using these estimates, the CPU runs Monte Carlo simulations to compute the expression of (43).

Information Interaction Process:

1) *Pilot Transmission:* UEs transmit pilot sequences to the CPU for initial channel estimation.
2) *Feedback Loop:* Once the channel is estimated, the CPU might send back an acknowledgment or even refined pilot sequences for the UEs to use in subsequent communications. This confirms the channel estimates are received and processed.
3) *Continuous Update:* In dynamic environments, the channel conditions can change rapidly, requiring the base station to periodically update the channel estimates. This can involve a repeated cycle of receiving new pilot sequences, running new Monte Carlo simulations, and reoptimizing the signal.

C. Power Control for Achievable Secrecy Rate of Downlink

In this section, a PC optimization problem is proposed to maximize the minimum ASR of all UEs, as given by

$$\text{PC:} \quad \max_{\{\eta_k:\, k\in\mathcal{K}\}} \ \min_{k\in\mathcal{K}} \ \text{ASR}_k^{dl} \tag{46}$$

$$\text{subject to:} \quad 0 \le \eta_k \le \eta_{\max} \ \forall k \in \mathcal{K}. \tag{46a}$$

We introduce some notations for SINRs at UEs as follows:

$$\begin{aligned} b_k^{\text{UE}} &\triangleq \left|\mathbb{E}\{\boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_k\}\right|^2 \in \mathbb{R}_+ \ \forall k \in \mathcal{K} \\ \mathbf{c}_k^{\text{UE}} &= \left(c_{k,1}^{\text{UE}} \ldots c_{k,K}^{\text{UE}}\right)^{\text{T}} \in \mathbb{R}_+^K \ \forall k \in \mathcal{K} \\ c_{k,i}^{\text{UE}} &= \begin{cases} \mathbb{E}\left\{\left|\boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_i\right|^2\right\}, & \text{if } k \ne i \\ \mathbb{E}\left\{\left|\boldsymbol{\mathcal{F}}_k^H \widehat{\boldsymbol{\mathcal{F}}}_k\right|^2\right\} - b_k^{\text{UE}}, & \text{if } k = i. \end{cases} \end{aligned} \tag{47}$$

Besides, we also define some notations for SINRs at Eves as follows:

$$\begin{aligned} b_k^{\mathscr{E}} &\triangleq \mathbb{E}\left\{\left|\boldsymbol{\mathcal{F}}_{\mathscr{E}_k}^H \widehat{\boldsymbol{\mathcal{F}}}_k\right|^2\right\} \in \mathbb{R}_+ \ \forall k \in \mathcal{J} \\ \mathbf{c}_k^{\mathscr{E}} &= \left(c_{k,1}^{\mathscr{E}} \ldots c_{k,K}^{\text{UE}}\right)^{\text{T}} \in \mathbb{R}_+^K \ \forall k \in \mathcal{J} \\ c_{k,i}^{\text{UE}} &= \begin{cases} \mathbb{E}\left\{\left|\boldsymbol{\mathcal{F}}_{\mathscr{E}_k}^H \widehat{\boldsymbol{\mathcal{F}}}_i\right|^2\right\}, & \text{if } k \ne i \\ 0, & \text{if } k = i. \end{cases} \end{aligned} \tag{48}$$

Hence, the SINRs (42) and (44) can be expressed as

$$\begin{aligned} \text{SINR}_k^{dl}(\boldsymbol{\eta}) &= \frac{b_k^{\text{UE}} \eta_k}{\mathbf{c}_k^{\text{UE}} \boldsymbol{\eta} + \sigma_{dl}^2} \ \forall k \in \mathcal{K} \\ \text{SINR}_{\mathscr{E}_k}^{dl}(\boldsymbol{\eta}) &= \frac{b_k^{\mathscr{E}} \eta_k}{\mathbf{c}_k^{\mathscr{E}} \boldsymbol{\eta} + \sigma_{dl}^2} \ \forall k \in \mathcal{J} \end{aligned} \tag{49}$$

where $\boldsymbol{\eta} \triangleq (\eta_1, \ldots, \eta_K)$.

Lemma 2: The (PC) problem can be solved using the Fixed-Point Algorithm 1, which converges to a suboptimal point.

Algorithm 1 Fixed-Point Algorithm for the PC Problem

1) Initialize iteration counter: $\kappa \leftarrow 0$. Initialize arbitrary positive power control coefficients: $\eta_k^{(\kappa)} > 0$, for $k = 1, \ldots, K$. $\Psi_i(\boldsymbol{\eta}) \triangleq \text{ASR}_k^{dl}(\boldsymbol{\eta})$.
2) Set a convergence threshold epsilon $\varepsilon = 0.0001$.
3) Repeat the following steps until convergence:
a) For each $\eta_i^{(\kappa)}$, updating it as follows:

$$\eta_i^{(\kappa+1)} = \Psi_i^{(\kappa)}(\boldsymbol{\eta}). \tag{51}$$

b) For each $\eta_i^{\kappa+1}$, scaling it as follows:

$$\eta_i^{(\kappa+1)} = \frac{\eta_{\max}}{\max\limits_i \eta_i^{(\kappa+1)}} \eta_i^{(\kappa+1)} \tag{52}$$

c) Calculate the maximum change in any η_i,

$$\Delta_x^{\kappa} = \max\{|\boldsymbol{\eta}_i^{\kappa+1} - \boldsymbol{\eta}_i^{\kappa}|\} \ \forall i \ = \ 1, 2, \ldots, K. \tag{53}$$

d) If $\Delta_x^{\kappa} < \varepsilon$, stop the iteration and output vector $\boldsymbol{\eta}^{\kappa+1}$.

Proof: The following inequalities are held:

$$\begin{aligned} 0 \le \text{SINR}_k^{dl}(\boldsymbol{\eta}) \le \text{SINR}_{\max}^{\text{UE}} &\triangleq \frac{b_k^{\text{UE}} \eta_{\max}}{\left(c_{k,k}^{\text{UE}} \eta_{\max} + \sigma_{dl}^2\right)} \\ 0 \le \text{SINR}_{\mathscr{E}_k}^{dl}(\boldsymbol{\eta}) \le \text{SINR}_{\max}^{\mathscr{E}} &\triangleq \frac{b_k^{\mathscr{E}} \eta_{\max}}{\left(c_{k,k}^{\mathscr{E}} \eta_{\max} + \sigma_{dl}^2\right)}. \end{aligned} \tag{50}$$

Consequently, the function ASR_k^{dl} is bounded and continuous on the constraint domain. It implies that the fixed-point algorithm converges to a suboptimal point. ■

D. Discussion on Joint Uplink and Downlink Transmission

Jointly considering uplink and downlink signal transmission in CF-mMIMO, as explored in [28], [29], and [30], opens avenues for optimized resource allocation, enhanced channel estimation, and improved interference management, which are pivotal benefits leading to better system performance and energy efficiency. However, this joint consideration also brings forth challenges such as increased system complexity, synchronization issues, and hardware limitations, which demand advanced algorithms and robust hardware support. In [28], the focus is on creating a model to switch between or jointly handle uplink and downlink data transmissions, aiming at optimizing spectral efficiency. Reference [29] ventures into secure transmission in downlink CF-mMIMO systems, hinting at a scenario where joint consideration of uplink and downlink phases could ensure secure transmissions. Meanwhile, [30] delves into a comprehensive optimization framework for transceiver design in network-assisted full-duplex (NAFD) systems to tackle cross-link interferences and maximize total uplink and downlink rates under various constraints. These studies unfold the intertwined narrative of benefits and challenges in jointly considering uplink and downlink signal transmission, showcasing the potential for enhancing spectral efficiency, managing interuser interference, and introducing novel solutions, such as joint sparse beamforming and PC.

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TABLE II
DNN MODEL

Layer	Dimension	Act. function
Input	$N^2(LK+2LRMK)$	
Layer 1	1024	relu
Layer 2	512	relu
Layer 3	256	relu
Output	$LKN+LK+K$	

This approach addresses the challenges and exploits the benefits inherent in joint uplink and downlink data transmission in modern communication systems.

V. DEEP LEARNING MODEL

In this section, we describe a DNN model designed to reduce the computation time required for obtaining $\mathbf{v}_{lk}^{\text{MMSE}}$ (36), optimal phase shift for PSS problem (28), and optimal transmit powers $\boldsymbol{\eta}^\star$ for PC problem (46), which involve high-complexity computations. It is important to note that these computations primarily rely on certain values, namely $\mathbf{R}_{lk}^{\mathbf{h}}$ in (3), $\mathbf{R}_{lrmk}^{\mathcal{G}}$ in (12), and $\mathbf{R}_{lrmk}^{\mathcal{F}}$ in (12). As a result, the DNN model can be designed as follows.

1) Data Set Design: We deploy a large number of different network topologies, varying in the number and location of UEs and APs. For each network topology, the CPU generates samples for the data set, where a sample $\mathcal{S}$ is generated as follows.

1) We consider the matrices $\mathbf{R}_{lk}^{\mathbf{h}}$, $\mathbf{R}_{lrmk}^{\mathcal{G}}$, and $\mathbf{R}_{lrmk}^{\mathcal{F}}$ as vectors with N^2 dimensions. It is important to normalize the vectors $\mathbf{R}_{lk}^{\mathbf{h}}$, $\mathbf{R}_{lrmk}^{\mathcal{G}}$, $\mathbf{R}_{lrmk}^{\mathcal{F}}$, $\mathbf{v}_{lk}^{\text{MMSE}}$, $\boldsymbol{\phi}^*$, and $\boldsymbol{\eta}^\star$ before adding them to the data set in order to improve the efficiency of training the DNN model.
2) *Input Parameters:* The input vector is designed by an $N^2(LK+2LRMK)$-dimensional vector
$$\left(\{\mathbf{R}_{lk}^{\mathbf{h}}\}, \{\mathbf{R}_{lrmk}^{\mathcal{G}}\}, \{\mathbf{R}_{lrmk}^{\mathcal{F}}\}\right), \text{ for all } l, r, m, k. \quad (54)$$
3) *Output parameters:* An output vector is designed by $(NLK+2K)-$dimensional vector
$$\left(\{\mathbf{v}_{lk}^{\text{MMSE}}\}, \boldsymbol{\phi}^*, \boldsymbol{\eta}^\star\right), \text{ for all } l, k. \quad (55)$$

2) DNN Model Design: The DNN model can be designed as in Table II.

In the DNN model, the stochastic gradient descent (SGD) optimization algorithm is used to compute and update the weights based on the learning rate during the backpropagation procedure. The loss function is considered as the mean-squared error (MSE) which is computed by taking the average of the squared differences between the predicted and actual values.

A. Deploying DNNs: Resource Constraints

Deploying a DNN in a MANET presents significant challenges in terms of resource constraints, such as energy consumption, time efficiency, and computational load. To mitigate these constraints, we use the following strategies.

1) *Centralized Solution:* We solve the PSC and PC problems at the CPU. This central approach enhances resource management and efficiency.

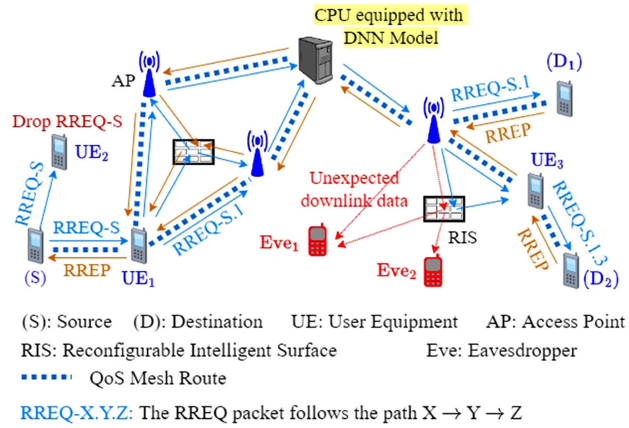

Fig. 6. Proposed QoS multicast routing protocol: QSMR protocol.

2) *High-Performance CPU:* We operate under the assumption that the CPU, akin to a supercomputer, can handle intense DNN computations and essentially possesses limitless energy. This premise sidesteps traditional energy constraints of DNN training, allowing a sharper focus on real-time tasks.
3) *Initial Training and Ongoing Updates:* The DNN starts with offline pretraining on robust computing clusters, capitalizing on past data. This model enters the MANET for live tasks. Recognizing the dynamics of MANET, the CPU with unlimited energy operates scheduled updates and refinements to the DNN model during low-activity intervals. This adaptability ensures the DNN consistently aligns with the MANET's changing landscape.

VI. PROPOSED QOS MULTICAST ROUTING PROTOCOL UTILIZING CROSS-LAYER DESIGN: QSMR PROTOCOL

In this section, we introduce the proposed QSMR protocol designed to establish the QMM route. This route originates from a source and extends to multiple destinations, leveraging information from both the physical and network layers. This information encompasses SEs, ASRs, position, direction, E2E delay, route stability, and hop count. Consequently, the QSMR protocol exhibits reduced routing delay, minimized control overhead, and an elevated PDR. A comprehensive, step-by-step breakdown of the proposed QSMR protocol is illustrated in Fig. 6. The subsequent parts of this section are structured as follows:

Section VI-A delves into direct and cascaded routing links. Section VI-B elucidates the QoS metrics. Section VI-C discusses the constraints of the QMM route. Finally, the intricacies of the proposed QSMR protocol are detailed in Section VI-E.

A. Direct and Cascaded Routing Link

Each link $(\text{UE}_i, \text{UE}_k)$ can be considered as the direct link $(\text{UE}_i, \text{UE}_k)^{\mathbf{dir}}$ or the cascaded link, where the cascaded link is defined as
$$(\text{UE}_i, \text{UE}_k)^{\mathbf{cas}} = (\text{UE}_i, \{\text{AP}\}, \mathbf{CPU}, \{\text{AP}\}, \text{UE}_k). \quad (56)$$

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B. QoS Metrics

1) End-to-End Delay: The time taken to send a packet from UE_i to UE_k is represented by $\mathrm{DL}(i,k)$, which is essentially the packet delay. By timestamping the packet before transmission, UE_k can obtain the packet delay $\mathrm{DL}(i,k)$, which can be calculated using the following equation:

$$\mathrm{DL}(i,k) = \mathrm{arriv}(k) - \mathrm{start}(i) \tag{57}$$

where $\mathrm{start}(i)$ is the time at which the packet is sent from UE_i and $\mathrm{arriv}(k)$ is the time at which the packet arrives at UE_k. The E2E delay of a route $\mathcal{R} \triangleq (\mathrm{UE}_1, \mathrm{UE}_2, \ldots, \mathrm{UE}_n)$ is defined by

$$\mathrm{DL}(\mathcal{R}) = \sum_{i=1}^{n-1} \mathrm{DL}(i, i+1). \tag{58}$$

Note that the E2E delay is computed for both direct and cascaded links.

2) Link Stability: At the time t, suppose that UE_i has the position $\mathbf{pos}_i(t) \triangleq (x_i(t), y_i(t))$, direction α_i and speed sp_i, and UE_k has the position $\mathbf{pos}_k(t) \triangleq (x_k(t), y_k(t))$, direction α_k and speed sp_k. The velocities of UE_i and UE_k at time t are given by the derivatives of their position vectors with respect to time

$$\begin{aligned} \mathbf{vel}_i(t) &= \big(\mathrm{sp}_i \cos(\alpha_i), \mathrm{sp}_i \sin(\alpha_i)\big) \\ \mathbf{vel}_k(t) &= \big(\mathrm{sp}_k \cos(\alpha_k), \mathrm{sp}_k \sin(\alpha_k)\big). \end{aligned} \tag{59}$$

To find the rate of change of the distance between UE_i and UE_k, we can use the square distance formula

$$\begin{aligned} \mathrm{sd}(t) &= \|\mathbf{pos}_i(t) - \mathbf{pos}_k(t)\|^2 \\ &= (x_i(t) - x_k(t))^2 + (y_i(t) - y_k(t))^2. \end{aligned} \tag{60}$$

Taking the derivative of $\mathrm{sd}(t)$ with respect to time t, we get

$$\begin{aligned} \mathrm{sd}'(t) &= \mathrm{d}(\mathrm{sd}(t))/\,\mathrm{d}\,t \\ &\quad + 2(y_i(t) - y_k(t))\big(\mathrm{sp}_i \sin(\alpha_i) - \mathrm{sp}_k \sin(\alpha_k)\big) \\ &= 2\langle \big(\mathbf{pos}_i(t) - \mathbf{pos}_k(t)\big), (\mathbf{vel}_i(t) - \mathbf{vel}_k(t))\rangle. \end{aligned} \tag{61}$$

The link stability for direct link $(\mathrm{UE}_i, \mathrm{UE}_k)^{\mathbf{dir}}$ is defined by

$$\mathrm{LS}(i,k) = 1 - \frac{|\,\mathrm{sd}'(t)|}{|\,\mathrm{sd}'(t)|_{\max}} \tag{62}$$

where $|\,\mathrm{sd}'(t)|_{\max}$ is computed by the absolute value of $\mathrm{sd}'(t)$ with $\mathrm{sp}_i = \mathrm{sp}_k = \mathrm{sp}_{\max}$, $\alpha_i = 0$ and $\alpha_k = 2\pi$. We have $0 \le \mathrm{LS} \le 1$. Moreover, if the LS is close to 1, the change in distance between UE_i and UE_k is very small, leading to high link stability. In RIS-aided CF-mMIMO, it is always possible to establish a link between any UE and some APs. As a result, the link stability of any cascaded link $(\mathrm{UE}_i, \mathrm{UE}_k)^{\mathbf{cas}}$ is set to 1.

3) Hop Count: Hop count of an $\mathcal{R}$ in (58) is computed by $\mathrm{HC}(\mathcal{R}) = n - 1$.

4) Spectrum Efficiency and Achievable Secrecy Rate: If a UE_k needs, it can receive the uplink SE and downlink ASR information from CPU, which serves as a QoS metrics denoted by $\mathrm{SE}^{ul}(k)$ and $\mathrm{ASR}^{dl}(k)$, respectively.

C. QoS Multicast Mesh Route Constraint

The QoS multicast mesh route (MMR) constraint is designed to enable the proposed QSMR protocol to establish QMM routes that satisfy QoS constraints for each single path $\mathcal{R} = (\mathrm{UE}_1, \mathrm{UE}_2, \ldots, \mathrm{UE}_n)$ from a source to multiple destinations. These constraints include: $\mathrm{SE}^{ul}(k) \ge \mathrm{SE}^{ul}_{\mathrm{th}}$ with $1 \le k < n$, $\mathrm{ASR}^{dl}(k) \ge \mathrm{ASR}^{dl}_{\mathrm{th}}$ with $1 < k \le n$, $\mathrm{DL}(\mathcal{R}) \le \mathrm{DL}_{\mathrm{th}}$, $\mathrm{LS}(k, k+1) \ge \mathrm{LS}_{\mathrm{th}}$ and $\mathrm{HC}(\mathcal{R}) \le \mathrm{HC}_{\mathrm{th}}$. Here, the notation th denotes the threshold.

D. Route Cost Vector

For a route $\mathcal{R} \triangleq (\mathrm{UE}_1, \ldots, \mathrm{UE}_n)$, the QoS cost function of $\mathcal{R}$ is defined as

$$\mathbf{cost}(\mathcal{R}) \triangleq \Big(\mathrm{SE}^{ul}(\mathcal{R}), \mathrm{ASR}^{dl}(\mathcal{R}), \mathrm{DL}(\mathcal{R}), \mathrm{LS}(\mathcal{R}), \mathrm{HC}(\mathcal{R})\Big) \tag{63}$$

where $\mathrm{DL}(\mathcal{R})$ is computed by (58)

$$\begin{aligned} \mathrm{SE}^{ul}(\mathcal{R}) &\triangleq \min\{\mathrm{SE}^{ul}(k) \colon 1 \le k < n\} \\ \mathrm{ASR}^{dl}(\mathcal{R}) &\triangleq \min\{\mathrm{ASR}^{dl}(k) \colon 1 < k \le n\} \\ \mathrm{LS}(\mathcal{R}) &\triangleq \min\{\mathrm{LS}(k, k+1) \colon 1 \le k < n\}. \end{aligned} \tag{64}$$

For two RCs $\mathcal{R}_1$ and $\mathcal{R}_2$, the inequality $\mathbf{cost}(\mathcal{R}_1) \le \mathbf{cost}(\mathcal{R}_2)$ is hold if only if

$$\begin{aligned} \mathrm{SE}^{ul}(\mathcal{R}_1) &\ge \mathrm{SE}^{ul}(\mathcal{R}_2) \text{ and} \\ \mathrm{ASR}^{dl}(\mathcal{R}_1) &\ge \mathrm{ASR}^{dl}(\mathcal{R}_2) \text{ and} \\ \mathrm{DL}(\mathcal{R}_1) &\le \mathrm{DL}(\mathcal{R}_2) \text{ and} \\ \mathrm{LS}(\mathcal{R}_1) &\ge \mathrm{LS}(\mathcal{R}_2) \text{ and} \\ \mathrm{HC}(\mathcal{R}(\mathcal{R}_1) &\le \mathrm{HC}(\mathcal{R}(\mathcal{R}_2). \end{aligned} \tag{65}$$

We define the RC threshold as

$$\mathbf{cost}_{\mathrm{th}} \triangleq \Big(\mathrm{SE}^{ul}_{\mathrm{th}}, \mathrm{ASR}^{dl}_{\mathrm{th}}, \mathrm{DL}_{\mathrm{th}}, \mathrm{LS}_{\mathrm{th}}, \mathrm{HC}_{\mathrm{th}}\Big). \tag{66}$$

E. Proposed QSMR Protocol

The proposed QSMR protocol is designed as an on-demand routing protocol under the random waypoint (RWP) mobility model. It incorporates techniques from the conventional AODV protocol [18] to handle packet processes related to sequence numbers. The QSMR protocol is designed to function within sessions, each consisting of processes including route discovery, RREP, data transmission, and route maintenance. We denote S as the source node, and $\mathcal{D} = \{\mathrm{D}_1, \ldots, \mathrm{D}_N\}$ as the multicast group. We assume that Multicast-Leader exists to represent the multicast group. This leader randomly chooses the source node for the multicast group. The QSMR protocol can be presented as follows.

Initialization:

1) By using cross-layer design in Section II-B and Fig. 5, UEs, APs, and CPU perform their tasks and calculations to determine the optimal phase shift of the RISs and transmit powers of APs that maximize the minimum uplink SEs and the downlink ASRs for all UEs, considering the presence of Eves. Moreover, to save computation time, the CPU utilizes a DNN model

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to predict combining vectors, optimal transmit power, SE, and ASR.

2) Each node can use the cross-layer design to obtain its movement information (movement_info) that includes the following fields:

$$[\textit{position, speed, direction}].$$

3) Before routing, if destination in $\mathfrak{D}$ needs to receive data from a source S, it will use the k-hops flooding method for new RREQ in [31, Sec. IV-B] to inform the $\mathfrak{D}$'s multicast IP address (*dst_IP_address*) to the S.
4) Each node initializes the routing table as follows:

$$\begin{bmatrix} \text{ROUTING_TABLE} \\ \textit{next_node, previous_node, route_cost} \end{bmatrix}$$

where *next_node* = null, *previous_node* = null and the terms of *route_cost* for $\mathcal{R}(\mathrm{S} \rightarrow \mathrm{S})$ in (63) is given by $\mathrm{SE}^{ul} = +\infty$, $\mathrm{ASR}^{dl} = +\infty$, $\mathrm{DL} = 0$, $\mathrm{LS} = 0$, and $\mathrm{HC} = 0$.

5) When the S receives the *dst_IP_address* and the data requirement from multicast group, the S will perform the MMR discovery process.

MMR Discovery Process:

1) *Step 1:* If S needs to establish an MMR to $\mathfrak{D}$, it generates an RREQ packet and broadcasts it to the neighbors. The RREQ packet includes the following fields:

$$\begin{bmatrix} \textit{packet_type=RREQ, link_type, rreq_id,} \\ \textit{sender_id, movement_info, last_visit, route_cost,} \\ \textit{src_IP_address, src_seq_number} \\ \textit{dst_IP_address, dst_seq_number, start_time} \end{bmatrix}$$

where *link_type* = 0 (0: direct link, 1: cascaded link), *sender_id* = S, *last_visit* = null, and the *route_cost* vector is obtained from the routing table. In Fig. 6, S broadcasts RREQ-S to neighbors.

2) *Step 2:* If an AP receives an RREQ packet from node v, it will process the following steps.
   a) AP forwards the RREQ packet to the CPU which then obtains all information from the RREQ packet.
   b) The CPU computes ASR^{dl} of all UEs to inform which node requires it.
   c) The CPU computes the $\mathrm{SE}^{ul}(v)$. CPU then updates the $\mathrm{SE}^{ul}(v)$ and *link_type* = 1 to the RREQ packet. Next, the CPU and APs broadcast the new RREQ packet to all UEs.

   In Fig. 6, APs receive RREQ-S.1 and forward them to CPU. Next, APs and CPU broadcast it to all UEs.
3) *Step 3:* If a node w receives an RREQ from CPU which is determined by *link_type* = 1, it will process the following.
   a) We consider $\mathcal{R}$ as the route from S to node w.
   b) The node w obtains all information from the RREQ packet.
   c) If node w is in last_visit, it drops the RREQ packet. This condition ensures that the RREQ is not received by a node more than once.
   d) It will require the information of $\mathrm{ASR}^{dl}(w)$ from CPU and update the information $\mathrm{ASR}^{dl}(w)$ to the RREQ's RC. It also updates the terms of RC from the RREQ as follows: $\mathrm{DL}_{\mathrm{new}}(\mathcal{R}) = \mathrm{DL}_{\mathrm{rreq}} + \mathrm{DL}(v, w)$, $\mathrm{LS}_{\mathrm{new}} = \min\{\mathrm{LS}_{\mathrm{rreq}}, \mathrm{LS}(v, w)\}$ and $\mathrm{HC}_{\mathrm{new}} = \mathrm{HC}_{\mathrm{rreq}} + 1$.
   e) If $\mathbf{cost}_{\mathrm{rreq}}(\mathcal{R}) > \mathbf{cost}_{\mathrm{th}}$ or $\mathbf{cost}_{\mathrm{rreq}}(\mathcal{R}) > \mathbf{cost}_{\mathrm{rt}}$, node w drops the RREQ, where $\mathbf{cost}_{\mathrm{rt}}$ is obtained from RC in the routing table.
   f) If $\mathbf{cost}_{\mathrm{rreq}}(\mathcal{R}) < \mathbf{cost}_{\mathrm{rt}}$, node w replaces $\mathbf{cost}_{\mathrm{rt}}$ in routing table by $\mathbf{cost}_{\mathrm{rreq}}(\mathcal{R})$. Node w then updates *previous_node* = v to the routing table.
   g) Node w generates and broadcasts the new RREQ by the same way as S in step 1.

   In Fig. 6, UE_1 and UE_2 receive the RREQ-S. We assume that UE_2 drops the RREQ-S because $\mathbf{cost}_{\mathrm{rreq}}(\mathcal{R}) > \mathbf{cost}_{\mathrm{th}}$. As a result, only UE_1 rebroadcasts the RREQ-S to its neighbors. Furthermore, D_1 and UE_3 receive RREQ-S.1. Subsequently, D_1 accepts RREQ-S.1 and proceeds to step 6. On the other hand, UE_3 generates and broadcasts new RREQs in the same manner as S did in step 1. Additionally, D_2 receives RREQ-S.1.3 from UE_3, leading D_2 to also proceed to step 6.
4) *Step 4:* If a node w receives an RREQ from a node v which is determined by *link_type* = 0, it will perform the same way as in step 3, but the information $\mathrm{ASR}^{dl}(w)$ is set by $+\infty$. In Fig. 6, $\mathrm{UE}_1, \mathrm{UE}_2$ receive RREQ-S. Then, they generate and broadcast new RREQs to neighbors.
5) *Step 5:* If node w is a destination in $\mathcal{D}$, it performs the RREP process.

RREP Process:

6) *Step 6:* If a D_i receives the RREQ in step 5, it will generate and reply an RREP packet to the previous node by unicast transmission. The RREP packet contains the following fields:

$$\begin{bmatrix} \textit{packet_type=RREP, sender_id, receiver_id, route_cost} \\ \textit{dst_IP_address, dst_seq_number, src_IP_address} \end{bmatrix}.$$

   Note that each destination D_i can accept multiple routes from different neighbors that satisfy the threshold condition, resulting in the establishment of a mesh route. In Fig. 6, D_1 and D_2 receive RREQs. Then, they generate and unicast RREPs to previous nodes.
7) *Step 7:* If a node receives an RREP packet, it performs the following tasks.
   a) The node adds *sender_id* to the *next_node* list in the routing table. If the node is not the S, it will reply the RREP packet to the previous hop in the same way as the D_i.
   b) If the source S receives all RREP from the multicast group $\mathfrak{D}$, it will perform the data transmission process.

   Otherwise, the node waits for an RREP packet until the RREP waiting time expires, it goes back to step 1. In Fig. 6, UE_3 and UE_1 receive RREPs. Then, they forward the RREPs to S.

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Data Transmission Process:

8) *Step 8:* Data packets are generated by the S based on a Poisson process. Then, the S transmits these data packets to the next nodes based on the routing table. When a node on the QMM route receives a data packet, it forwards the data packet to the next nodes in the same manner as the S. As a result, the intermediate nodes forward the data packets to the D based on the QMM route.

Broken Link Detection and Route Maintenance:

1) In this article, the methods for detecting broken links and maintaining routes are implemented in the same manner as described in [31].

F. Implementing QSMR: Integration of Algorithms and Techniques in Cross-Layer Design

The proposed QSMR protocol handles the processes of both the physical and network layers to ensure smooth and secure communication flows. The integration of algorithms and techniques is presented as follows.

Physical Layer Activities: Physical layer tasks generally serve as the starting point and occur continuously. When a new UE wishes to join the network, it sends a pilot signal to the APs. The physical layer performs the following tasks.

1) The CPU computes channel estimates as outlined in Section III.
2) The CPU solves the PSS problem, described by (28), to select the optimal phase shifts for the RISs to maximize the minimum strength of aggregate channel estimates from UEs to APs through RISs.
3) The CPU calculates the optimal MMSE combining using (36) to maximize the uplink SEs for the UEs.
4) The CPU determines the optimal transmit powers for the APs to maximize the minimum ASRs for all UEs using Algorithm 1.
5) UEs utilize GPS technology to ascertain their positions and velocities.

All these steps serve as critical precursors, setting the stage for the network layer to execute its functions.

Network Layer Operations: Once the physical layer has secured a stable connection and calculated important QoS metrics, the network layer takes over.

1) The UEs retrieve information on SE, ASR, position, and velocity from the physical layer. Additionally, they gather data on E2E delay, route stability, and hop count from the network layer itself.
2) The collected metrics are employed in the proposed QSMR protocol, as detailed in Section VI, to establish QMM routes.

Cross-Layer Synergy: The system also features an integrated cross-layer approach, allowing for information to be cycled back and forth between the layers for dynamic optimization. For example, if the network layer needs to re-establish new QMM routes, this requirement can be relayed to the physical layer for fine-tuning of phase shifts or power allocations, and for updating new QoS metrics.

TABLE III
PARAMETERS FOR PERFORMANCE EVALUATION

Network size	$1{,}000 \times 1{,}000\ \mathrm{m}^2$
Number of nodes	50
Mobility model	Random Way Point
Max speed of nodes	20 km/h $\sim$ 80 km/h
Transmission range of nodes	250 m
Data transmission process	Poisson
Data transmission rate	5 packets/s
Control packet size	576 bits
Data packet size	512 bytes
Session length	10 s
Simulation time	$1,000$ s

In summary, although initiation often starts with the physical layer, both layers engage in continuous interplay, particularly in a dynamically changing environment like that of ours that utilizes cross-layer design. This dual-layer cooperation enables our system to meet evolving demands and challenges efficiently.

VII. PERFORMANCE EVALUATION

Based on the deployment strategies in [11], [23], and [32], we deploy an RIS-aided CF-mMIMO system, including $L = 80$ APs with four antennas, $R = 16$ RISs with nine elements, $K = 50$ single-antenna IoT devices (UEs), and $J = 10$ active Eves that are independently and uniformly distributed within a square of size $1 \times 1\ \mathrm{km}^2$. The distance between any (UE and AP), (UE and RIS), and (AP and RIS) is larger than $35\ \mathrm{m}$. The antenna spacing and reflecting element spacing are both $1/2$. We set the pilot power as 100 mW, the maximum UL transmit power as 1 W, and 20-MHz bandwidth. The Gaussian local scattering model is under consideration, where the scattering components are arranged in a Gaussian manner around the nominal azimuth and elevation angles, represented by φ and θ, respectively. The deviation in the azimuth and elevation angles are denoted by δ and ε, respectively, and are independently distributed according to normal distributions $\delta \sim \mathcal{N}(0, \sigma_\varphi^2)$ and $\varepsilon \sim \mathcal{N}(0, \sigma_\theta^2)$, respectively [23]. The (i, j)th element of normalized spatial correlation matrices of channels between UEs, APs, and RISs is computed by [23, Lemma 6] with $\sigma_\varphi^2 = \sigma_\theta^2 = \pi/12$. The coherence block length is $\tau_c = 10\,000$ samples, and $\tau_p = (RM + 2)K$ mutually orthogonal pilot sequences. The large-scale fading coefficient for the LoS path is modeled (in dB) by

$$\beta_{lk} = -30.18 - 26\log_{10}(d_{lk}) + \mathcal{X}_{lk} \tag{67}$$

where $\mathcal{X} \sim \mathcal{N}(0, \sigma_{\mathrm{sf}}^2)$ is the shadow fading with $\sigma_{\mathrm{sf}}^2 = 4$. The thresholds for the routing cost vectors are defined as $\mathrm{SE}_{\mathrm{th}}^{ul} = 2 \times 10^{-4}$, $\mathrm{ASR}_{\mathrm{th}}^{dl} = 10^{-4}$, $\mathrm{DL}_{\mathrm{th}} = 1s$, $\mathrm{LS}_{\mathrm{th}} = 0.5$, and $\mathrm{HC}_{\mathrm{th}} = 5$. The environments and other parameters for the routing protocol are collected from [33], [34], and [35], as shown in Table III.

The proposed QSMR is evaluated based on the following metrics.

1) The routing delay is defined as the average time taken to establish multicast routing routes (mesh route) per session.

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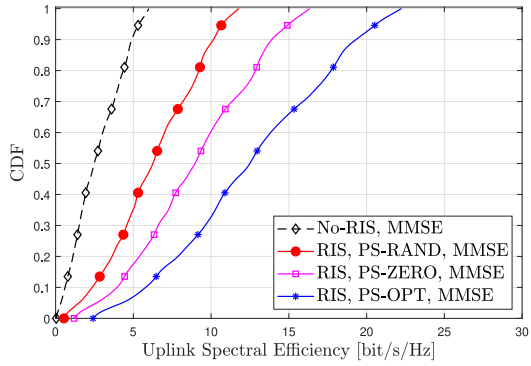

Fig. 7. CDF of SEs under different phase-shifts configuration.

2) The control overheads are defined as the average number of control packets per node per session required to establish multicast routing routes (mesh route).
3) The PDR is defined as the ratio of the number of data packets received at the destination nodes (multicast member nodes) to the number of data packets sent from the source per session.

For the simulation results, we use the following abbreviations.

1) *PS-OPT* denotes the optimal phase shifts of RISs. It is the solution to the PSS problem (28), which can be solved by CVXPY. The PS-OPT maximizes the minimum strength of aggregate channel estimates for UEs.
2) *POW-OPT* denotes the optimal transmit power for APs. It is the solution to the (PC) problem (46), which can be solved by Algorithm 1. The POW-OPT maximizes the minimum ASR for UEs.
3) *CLAF* denotes the CLAF-AODV [16]. The CLAF protocol is modified to establish the multicast tree structure based on the QoS metrics, such as delay, throughput, packet loss, stability, and adaptability of the network.
4) *HGFA* denotes the hybrid genetic firefly algorithm-based routing protocol in [17]. The HGFA is modified to establish the multicast tree structure based on the fitness function including the retransmission time and propagation time parameters.

Fig. 7 shows the cumulative distribution function (CDF) of the uplink SE for both the RIS-aided CF-mMIMO system and the conventional CF-mMIMO system without RIS (No-RIS), using MMSE combining under different phase-shift configurations. The results indicate that the uplink SE achieved with the optimal phase-shift configuration (PS-OPT) outperforms that achieved with a zero phase-shift (PS-0) and a random phase-shift (PS-RAND). The reason is that the PS-OPT can maximize the minimum strength of aggregate channel estimates for UEs. In contrast, the high interference caused by numerous RISs results in the lowest SE under the PS-RAND configuration. Moreover, the uplink SEs are also maximized by (37) and (38), showing significant improvement in the proposed system. These results suggest that the presence of RISs can substantially enhance the uplink SEs of the proposed system.

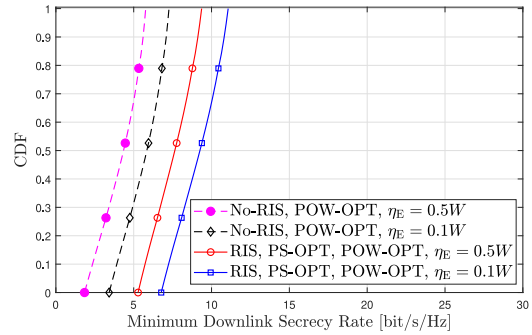

Fig. 8. CDF of the minimum downlink ASR.

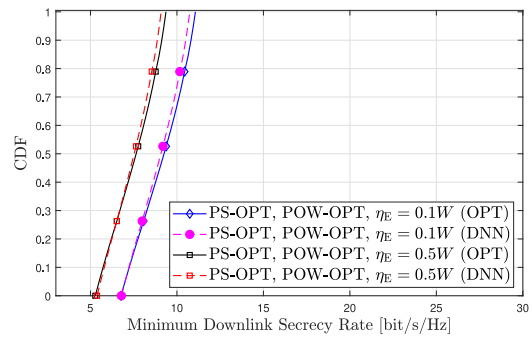

Fig. 9. Comparison of optimization and DNN methods for solving (PSS) and (PC) problems.

Fig. 8 presents the CDF of the minimum downlink ASR for both the RIS-aided CF-mMIMO system and the conventional CF-mMIMO system under different transmit power levels of the active Eves (η_E). The results show that when PS-OPT and POW-OPT are applied, the RIS-aided CF-mMIMO system achieves the highest minimum ASR at a η_E setting of 0.1 W, leading to a decrease in ASR for all systems. The reason is that an increase in Eve's transmit power introduces greater interference into the network, affecting signal quality and thereby reducing the ASR. Additionally, PS-OPT strengthens the signal between APs and legitimate UEs in the RIS-aided CF-MIMO system. Thus, the combination of PS-OPT and POW-OPT leads to improved ASR compared to the conventional CF-mMIMO system.

Fig. 9 presents the CDF of the minimum downlink ASR for the RIS-aided CF-MIMO system using the optimization (OPT) and DNN methods. The OPT method is the combination of the PS-OPT and the POW-OPT, while the DNN method applies a DNN model to solve the (PSS) and (PC) problems. The similarity in the CDF functions for minimum ASR indicates the effectiveness of both methods in optimizing phase shifts and transmit powers.

The loss functions for the training and validation sets, which contain 5000 samples, are depicted in Fig. 10. As the number of training epochs increases, the loss function decreases and converges to 0.016. The average loss of the output vector is approximately 0.016, meeting the expected approximation for normalized output vectors. Although the loss may be high for very small original values, it is very accurate for important values when recovering values from predicted normalized

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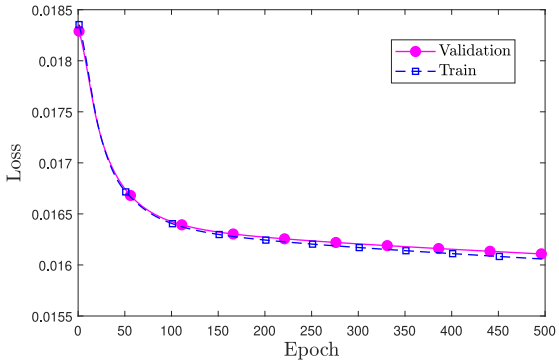

Fig. 10. Loss function for DNN model.

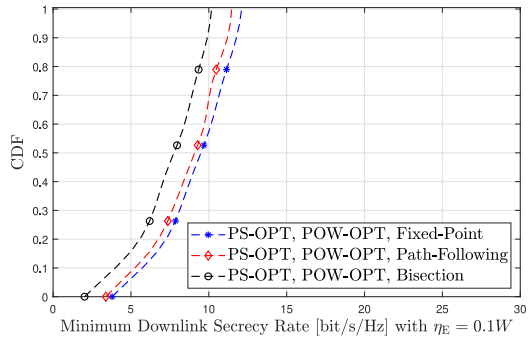

Fig. 12. CDF of the minimum downlink ASR with different methods.

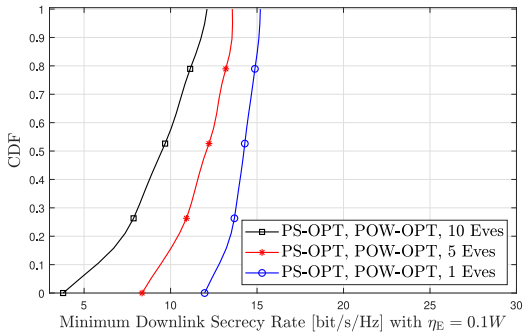

Fig. 11. CDF of the minimum downlink ASR with different number of Eves.

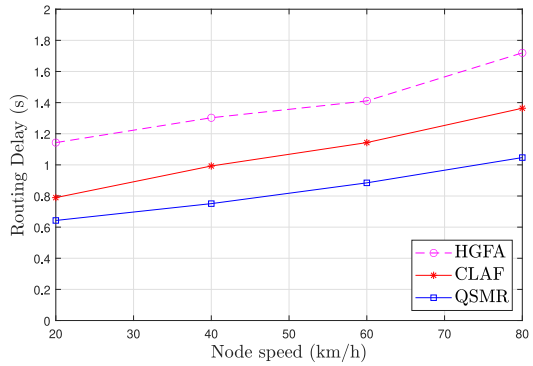

Fig. 13. Routing delay as a function of node speed.

output vectors. Moreover, the small values of the output vectors do not affect the SEs and ASRs. Therefore, using the DNN model produces CDF functions that are similar to those obtained by the OPT method as shown in Fig. 9. Additionally, the DNN's computation time is significantly faster than the simulation algorithm's, taking only 0.358502 s compared to 3.8284 s, respectively. Both programs were executed on a computer with an 11th Gen Intel Core i7-11700 @ 2.50-GHz 2.50-GHz processor.

Fig. 11 presents the minimum downlink ASR for the RIS-aided CF-MIMO system with different numbers of Eves. It can be observed that when the number of Eves increases, the minimum ASR decreases. The reason is that as the number of Eves increases, the APs might allocate resources less optimally during the max–min optimization process, negatively affecting the ASR of legitimate UEs. Moreover, the optimal transmit power calculated for the APs may shift toward mitigating potential security threats rather than maximizing the ASR, which can result in a decrease in the minimum ASR.

Fig. 12 illustrates the CDF of the minimum downlink ASR in the RIS-aided CF-MIMO system, comparing different methods, such as Fixed-Point, Path-Following, and Bisection. The Fixed-Point method, employed in our paper, focuses on solving the max–min fairness PC problem to maximize the minimum ASR. Previous studies have utilized the Bisection method [7] and the Path-Following method [11] to optimize ASR for a single Eve. Our observations indicate that the Fixed-Point method consistently yields the highest ASR compared to the other techniques. Additionally, the Path-Following method is more effective than the Bisection method. This superior performance of the Fixed-Point method can be attributed to its versatility in solving a range of problems, including game theory equilibriums, network flows, and specific types of differential equations, making it particularly effective for max–min fairness problems. The lower performance of the Bisection method is consistent with its limitations in accuracy, especially in cases that are either complex or high dimensional. Thus, it results in the lowest ASR among the three methods examined.

In Fig. 13, we illustrate the routing delay as a function of node speed for various routing protocols. It is observed that HGFA has the highest routing delay, while the proposed QSMR exhibits the lowest delay across the entire speed range. The primary reason for HGFA's elevated delay is its lack of consideration for SE, ASR, delay, route stability, and other QoS metrics. Furthermore, HGFA aims to establish a tree structure, which is more prone to breaking compared to a mesh route that takes QoS metrics into account. Consequently, HGFA consumes more time in both establishing and repairing routes, leading to higher routing delays than those seen in CLAF and the proposed QSMR. Although CLAF does consider QoS metrics, it also aims to establish a tree structure. In contrast, the proposed QSMR utilizes QoS metrics and optimal SE and ASR from a cross-layer design to establish a more stable and high-quality mesh route. This eliminates the need for time-consuming route repairs due to broken links, resulting in a lower routing delay compared to CLAF.

In Fig. 14, we examine the control overhead as a function of node speed under different routing protocols. We find that HGFA incurs the highest control overhead, while our proposed QSMR has the lowest overhead throughout the speed

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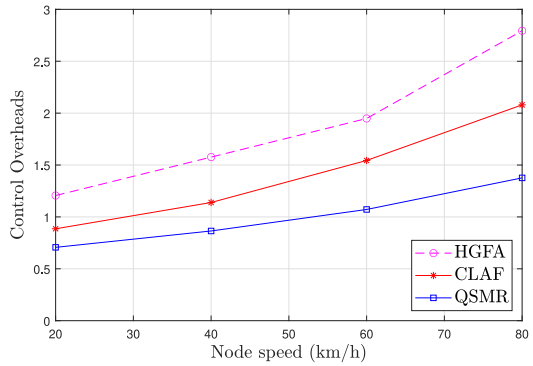

Fig. 14. Control overhead as a function of node speed per session per node.

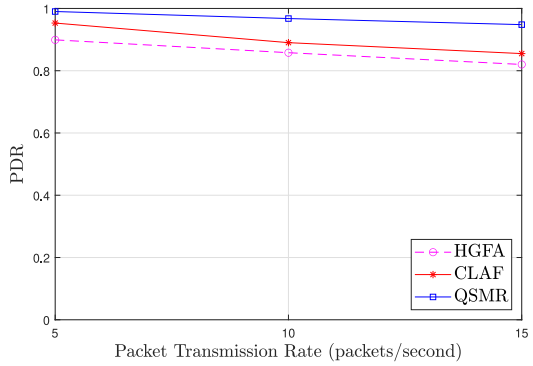

Fig. 15. PDR as a function of node speed at a fixed data packet transmission rate of five packets per second.

range. A major factor contributing to HGFA's higher control overhead is its simplistic design, which does not account for SE, ASR, delay, route stability, or other QoS metrics. As HGFA aims to build a tree structure, this requires frequent control messages for maintenance, especially when nodes or links fail. This naturally increases the control overhead. CLAF, although considering QoS metrics, also focuses on a tree structure, leading to a moderate level of control overhead. On the other hand, our proposed QSMR leverages optimal SE and ASR obtained through cross-layer design, as well as other QoS metrics, to create a more robust and stable mesh network. This significantly reduces the frequency of control messages needed for route maintenance and node coordination. As a result, QSMR demonstrates a substantially lower control overhead compared to both HGFA and CLAF.

In Fig. 15, we analyze the PDR as a function of node speed for different routing protocols. Our observations indicate that HGFA has the lowest PDR, while the proposed QSMR showcases the highest PDR across all speeds. One of the key reasons for HGFA's poor PDR performance is its neglect of essential factors, such as SE, ASR, delay, route stability, and other QoS metrics. Because HGFA aims to establish a tree structure, it is inherently more susceptible to link failures and route breaks, which in turn leads to a higher rate of packet loss and thus a lower PDR. While CLAF does consider QoS metrics, its focus on a tree-based structure also makes it vulnerable to link failures, albeit to a lesser extent than HGFA. This results in a moderate PDR compared to the

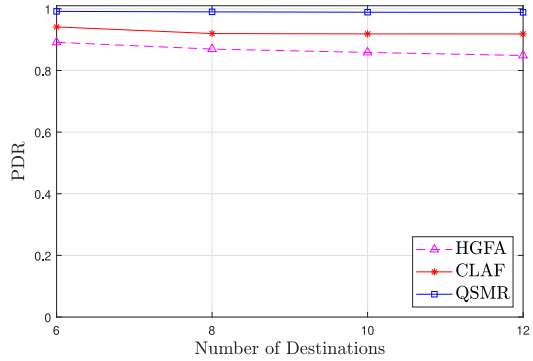

Fig. 16. Scalability as a function of number of destinations with a fixed node speed of 50 km/h.

other protocols. In contrast, the proposed QSMR integrates QoS metrics as well as optimal SE and ASR values obtained from a cross-layer design to establish a stable and robust mesh network. This design choice minimizes route failures and enhances route reliability, ensuring a higher rate of successful packet deliveries. As a result, QSMR achieves a significantly higher PDR compared to both HGFA and CLAF.

In Fig. 16, we examine the scalability of various protocols by plotting the PDR as a function of the number of multicast member nodes. Interestingly, it is observed that the PDR remains largely constant across all protocols, irrespective of the increasing number of destinations. This stable performance can be attributed to the underlying network architectures of the protocols, which are designed to establish either a tree or a mesh route for multicast data dissemination. The key advantage of these structures is their ability to deliver data packets to multiple destinations in a single transmission cycle. This eliminates the need for individual packet transmissions to each destination, thereby significantly reducing both transmission delay and network congestion. As a consequence, all protocols manage to maintain a consistent and stable PDR, even as the number of multicast member nodes increases. This observation underlines the scalability and efficiency of these routing protocols in handling multicast traffic.

VIII. Conclusion

This article aims to address the limitations of previous studies by proposing a novel QSMR protocol for IoT-enabled MANETs in RIS-aided CF-mMIMO systems with spatially correlated fading channels, multiple antennas APs, and multiple Eves. The proposed protocol satisfies QoS requirements, such as SE, ASRs, E2E delay, route stability, and hop count requirements. To achieve this, we first derive closed-form expressions for channel estimates and SEs for IoT devices in both uplink and downlink as well as the expressions of downlink ASRs for attacked IoT devices under the MMSE combining method. We then propose two novel max–min fairness optimization problems for IoT devices: a PSC problem to maximize the minimum uplink SEs, and a PC problem to maximize the minimum downlink ASRs. To reduce the computation time required to solve these problems, we design a DNN. Next, we propose the QSMR protocol

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that establishes a QMM route from a source to multiple destinations using information from both the physical and network layers, including SEs, ASRs, position, direction, E2E delay, route stability, and hop count. Finally, simulation results evaluate the secure performance and make comparisons for different scenarios, demonstrating that the proposed QSMR protocol outperforms other routing protocols in terms of routing delay, control overhead, and PDR in the presence of the Eves. Overall, this article contributes to enhance the efficiency, security, and reliability of multicast communication in the 6G and IoT networks by proposing a novel QSMR protocol that considers the RIS-aided CF-mMIMO system with spatially correlated fading channels, multiple antennas APs, and multiple Eves.

Future work includes extending the proposed QSMR protocol to incorporate unmanned aerial vehicles (UAVs) and vehicular networks in RIS-aided CF-mMIMO systems, as efficient multicast routing protocols are crucial for supporting various applications, such as infotainment and emergency communication. Additionally, we will address problems such as the power-efficient deployment of RISs in Remark 2 and joint uplink and downlink data transmission in Section IV-D. Finally, the QSMR protocol can be optimized to meet the diverse QoS requirements of 6G networks, which are expected to offer higher data rates, ultralow latency, and massive connectivity.

Appendix

Proof of Lemma 1

Based on (6), the correlation matrix of channel $\mathbf{g}_{lrm}$ over the mth element of RIS_r and AP_l is computed by

$$\begin{aligned}\mathbf{R}^{\mathbf{g}}_{lrm} &= \mathbb{E}\{\mathbf{g}_{lrm}\mathbf{g}^H_{lrm}\} = \mathbb{E}\{[\mathbf{G}_{lr}]_{:m}[\mathbf{G}_{lr}]^H_{:m}\}\\ &= \mathbb{E}\{\mathbf{A}_{lr}\mathbf{W}_{lr}[\mathbf{B}_{lr}]_{:m}[\mathbf{B}_{lr}]^H_{:m}\mathbf{W}^H_{lr}\mathbf{A}^H_{lr}\}.\end{aligned} \tag{68}$$

where

$$\begin{aligned}\mathbf{W}_{lr}[\mathbf{B}_{lr}]_{:m} &= \sum_{i=1}^{M}[\mathbf{W}_{lr}]_{:i}[\mathbf{B}_{lr}]_{im}\\ [\mathbf{B}_{lr}]^H_{:m}\mathbf{W}^H_{lr} &= \sum_{i'=1}^{M}[\mathbf{B}_{lr}]^*_{i'm}[\mathbf{W}_{lr}]^H_{:i'}.\end{aligned} \tag{69}$$

Hence, we have

$$\begin{aligned}\mathbf{R}^{\mathbf{g}}_{lrm} &= \sum_{i=1}^{M}\sum_{i'=1}^{M}[\mathbf{B}_{lr}]_{im}[\mathbf{B}_{lr}]^*_{i'm}\mathbb{E}\{\mathbf{A}_{lr}[\mathbf{W}_{lr}]_{:i}[\mathbf{W}_{lr}]^H_{:i'}\mathbf{A}^H_{lr}\}\\ &= \sum_{i=1}^{M}[\mathbf{B}_{lr}]_{im}[\mathbf{B}_{lr}]^*_{im}\mathbb{E}\{\mathbf{A}_{lr}\mathbf{A}^H_{lr}\} = [\mathbf{T}_{lr}]_{mm}\mathbf{S}_{lr}.\end{aligned} \tag{70}$$

According to (7) and (70), the correlation matrix of the cascaded channel $\mathcal{G}_{0,lrmk}$ between UE_k and AP_l through the mth element of RIS_r is computed by

$$\begin{aligned}\mathbf{R}^{\mathcal{G}}_{lrmk} &\triangleq \mathbb{E}\{\mathcal{G}_{0,lrmk}\mathcal{G}^H_{0,lrmk}\}\\ &= \mathbb{E}\{\mathbf{g}_{lrm}\mathbf{g}^H_{lrm}\}\mathbb{E}\{f_{rmk}f^*_{rmk}\} = \mathbf{R}^{\mathbf{g}}_{lrm}\left[\mathbf{R}^{\mathbf{f}}_{rk}\right]_{mm}.\end{aligned} \tag{71}$$

Thus, we can obtain the correlation matrix of the aggregate channel $\mathcal{F}_{0,lrmk}$ between UE_k and AP_l through the mth element of RIS_r from (9) and (71) as follows:

$$\begin{aligned}\mathbf{R}^{\mathcal{F}}_{lrmk} &\triangleq \mathbb{E}\{\mathcal{F}_{0,lrmk}\mathcal{F}^H_{0,lrmk}\} = \mathbf{R}^{\mathbf{h}}_{lk} + \mathbf{R}^{\mathcal{G}}_{lrmk}\\ &= \mathbf{R}^{\mathbf{h}}_{lk} + \mathbf{R}^{\mathbf{g}}_{lrm}\left[\mathbf{R}^{\mathbf{f}}_{rk}\right]_{mm}.\end{aligned} \tag{72}$$

Moreover, it is easy to obtain that the channels $\mathbf{g}_{lrm}$, $\mathcal{G}_{0,lrmk}$, and $\mathcal{F}_{0,lrmk}$ have the expected values of zero.

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